

ENTERPRISE NETWORKS

Introduction

- * When a person is at home, all calls relating to interpersonal and interactive application must be made using a PSTN or ISDN or a cable distribution network.
- * For an enterprise that occupies just a single site / establishment - for example, a small to medium sized company, a hospital complex, a university campus - the PBX and LAN at the establishment handle all internal calls and only those calls that are external to the site are made using an appropriate public network. Such as PSTN, an ISDN or an Internet service provider network. For an enterprise that operates over multiple sites, however, when there is a significant proportion of intersite calls, an alternative solution is to extend the private facilities associated with each site to embrace all sites. This involves linking the sites together using high bit rate transmission lines that are leased from a (national) network provider. The resulting network is then known as a multisite Enterprise network.
- * If the sites are located in different countries around the world, a global enterprise network.
- * Normally, leased lines are charged for on an annual rather than a per-call basis and hence this approach is only justified when the annual (public) call charges to other sites exceeds the cost of leasing the lines.
- * An added benefit of creating a private network, however, is that it is often easier to offer more sophisticated services since the utilization of the bandwidth of the intersite leased lines is under the control ~~(of the bandwidth of the intersite leased lines is under the)~~ of the enterprise network manager.

LANs

LANs are used to interconnected distributed communities of end systems - often referred to as stations

LANs includes - Multimedia PCs, workstations, servers, and so on. Typically these are distributed around an office, a single building or a localized group of building, all of which belong to a single enterprise.

- * operate using a shared, high bit rate, transmission medium to which all the stations are attached and the information frames relating to all calls are transmitted.
- * To ensure the transmission bandwidth is shared fairly between all of the attached stations.
- * There are two methods are used for a number of different medium access control (MAC)
 - 1] CSMA/CD [carrier-sense multiple-access with collision detection]
 - 2] Token passing
- * both of which have a defined maximum number of attached stations and length of transmission medium associated with them.
- * The maximum distance is relatively small and hence most LANs of this type comprise multiple segments that are interconnected together using either repeaters or devices known as bridges.
- * High bit rate (site-wide) backbone subnetworks

Ethernet / IEEE 802.3

- * Ethernet / IEEE 802.3 are used extensively in technical and office environments.
- * Ethernet has gone through many phases of development since its first introduction but, in general, the same basic mode of operation is still used.
- * All frame transmissions between all the stations that are attached to the LAN take place over a shared 10 Mbps bus and ~~CSMA~~ CSMA / CD MAC method.
- * Then used to share the use of the bus in an equitable way.

CSMA / CD

Since all the stations are attached directly to the same cable / bus, it is said to operate in a multiple access (MA) mode. To transmit a block of data.

- * The source station and its own address in the frame header and an FCS field at the tail of the frame.
- * The bus operates in the broadcast mode which means that every frame transmitted is received by all the other stations that are attached to bus. Hence as each of the other station receives the frame.
- * It first checks the frame is free of errors using the FCS and, if it is it compares the destination address in the header with its own address.
- * If they are different, the station simply discards the frame. If they are same, the frame contents are passed up to the LC sublayer for processing together with the address of the source station.
- * With this mode of operation, two or more stations may attempt to transmit a frame over the bus at the same time. Because of the broadcast mode, this will result in the contents of the two (or more) frames being corrupted and a collision is said to have occurred.

Hence in order to reduce the possibility of a collision, prior to sending a frame the source station first determine whether a signal/frame is currently being transmitted on the bus. If a signal - known as the carrier - is sensed.

- * The station defers its own transmission until the current frame transmission is complete and only then does it attempt to send its own frame.
- * Both will start to transmit their frame simultaneously on detecting that the transmission of the current frame is complete.
- * When this happens, however, it is necessary for the two (or more) station involved, to detect a collision has occurred before each has finished transmitting its own frame.
- * The possibly large signal propagation delay of the bus and the high transmission bit rate used (10Mbps). This is not as straightforward as it might seem.
- * A station detects that a collision has occurred by simultaneously monitoring the signal that is present on the cable all the time it is transmitting frame.
- * Then, if the transmitted and monitored signals are different, a collision is assumed to have occurred - collision detected (CD).
- * A station can experience a collision not just at the start of a frame but after it has transmitted a number of bits.
- * The worst case time delay - and hence maximum number of bits that have been transmitted before detecting that a collision has taken place is known as the collision window.
- * And occurs when the two collision has taken at stations are attached to opposite extremities of the bus.

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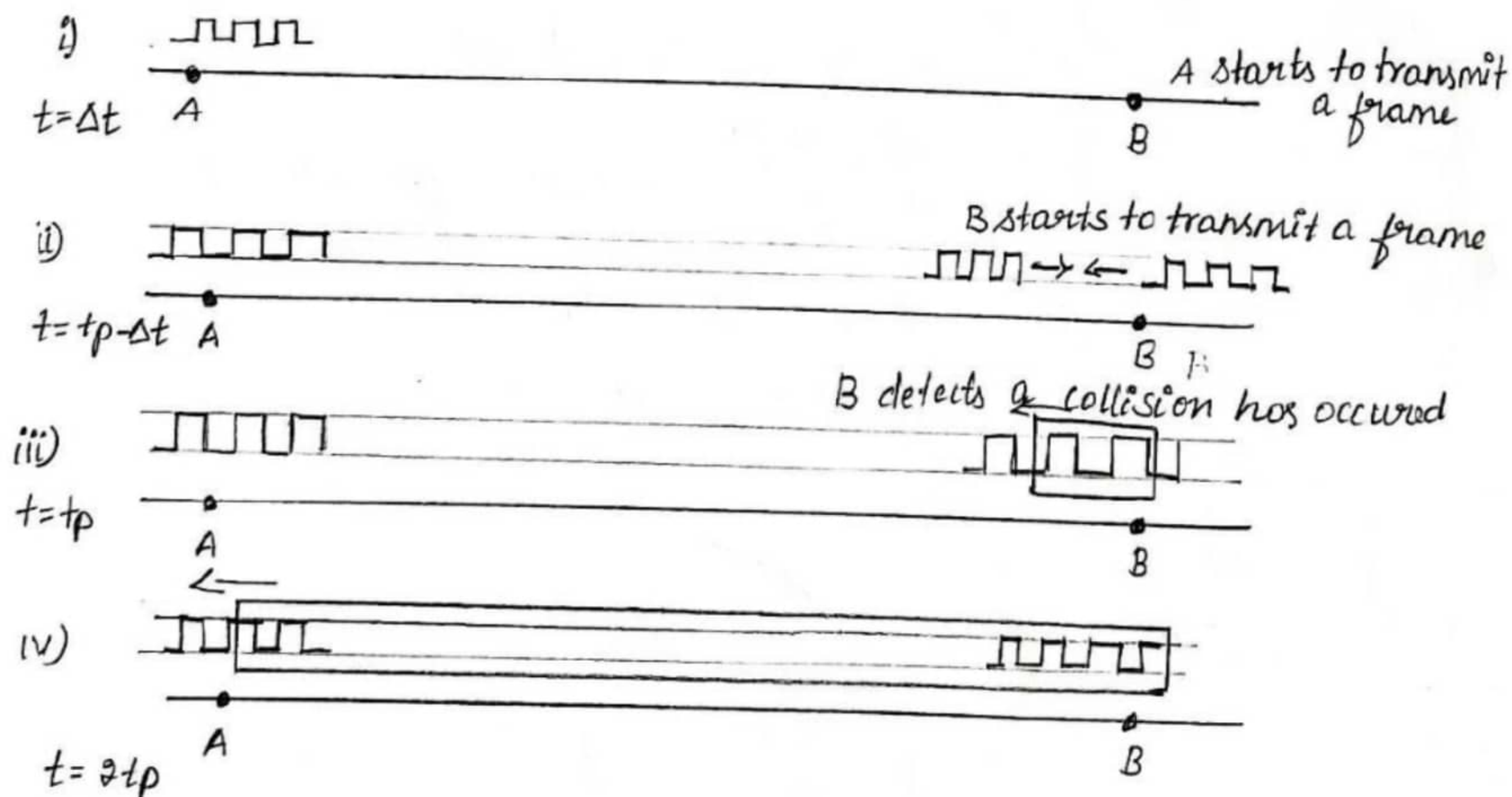
- * Station A has determined that no transmission is in progress and hence starts to transmit a frame.
- * In respect of the bit rate being used, the first bit of the frame will take a small but finite time to propagate over the transmission medium determined by the length of the cable 'L'.
- * And the signal propagation velocity $v \text{ ms}^{-1}$.
- * The maximum length of cable is set at 8.5 km, hence assuming a v of $2 \times 10^8 \text{ ms}^{-1}$. The worst case signal propagation delay time T_p going from one end of the cable to the other is given by

$$T_p = L/v$$

$$T_p = \frac{8.5 \times 10^3}{2 \times 10^8}$$

$$T_p = 19.5 \text{ microseconds.}$$

- * Station B determines the transmission medium is free and starts to transmit a frame.



$t_p =$ (worst case) transmission propagation (path) delay

CSMA/CD Worst-case collision detection

- * After B has transmitted just a few bits, the two signals collide and the collision signal then continues to propagate back to station A.
- * Hence the worst case time before station A detects that a collision has occurred, $9\mu s$ is $95\mu s$.
- * In Addition, as we shall expand upon later, in order to transmit the signal over this length of cable the cable is made up of five 500m segments.
- * All interconnected together by means of four device called repeaters.
- * Each repeater introduces a delay of a few microseconds in order to synchronize to each new frame.
- * Hence the total worst-case time is set at 50 microseconds or, assuming a bit rate of 10Mbps, after A has transmitted.

$$10 \times 10^6 \times 50 \times 10^{-6} = 500 \text{ bits.}$$

- * A safety margin of 12 bits is then added to this and the minimum frame size is set at 512 bits or 64 bytes / octets. This is called "Slot Time".
- * And ensure that station A will have detected a collision before it has transmitted its smallest frame.
- * to ensure that the collision signal persists for sufficient time for it to be detected by A. on detecting the collision, B continues to send a random bit pattern for a short period. This is known as the Jam sequence and is equal to 32 bits.

$$\underline{1518 \times 8}$$

$$1518 \times 8 / 10 \times 10^6 = 1.2144 \text{ milliseconds.}$$

clearly, the probability of this occurring increases with the level of traffic (number of frames) being generated and the maximum throughput of the LAN occurs when this limit is reached.

Hence if a second collision should occur when a station is trying to send a frame, this is taken as a sign that the cable is currently overloaded.

* To avoid further loading the cable, therefore, the time interval between trying to retransmit a frame is increased exponentially after each new attempt is made using a process known as truncated binary exponential backoff.

* The actual time is a function of the slot time

* The number of slot times before the N th retransmission attempt is chosen as a uniformly distributed random integer R in the range $0 \leq R < 2^k$, where $k = \min(N, \text{back off limit})$.

In the standard, the back off limit is set to 10.

Wiring Configuration:-

There are a number of different types of cables that can be used with Ethernet.

These includes:

* 10 Base 2

* 10 Base 5

* 10 Base T

* 10 Base F

* 10Base9

Thin wire (0.25 inch diameter) coaxial cable with a maximum segment length of 900m.

* 10Base5

Thick wire (0.5 inch diameter) coaxial cable with a maximum segment length of 500m.

* 10BaseT

Hub (star) topology with twisted pair drop cables of up to 100m.

* 10BaseF

Hub (star) topology with optical fiber drop cables of up to 1.5km

Although different types of cable are used, they all operate using the same CSMA/CD MAC method.

* At the time the first Ethernet installations were carried out, the only transmission medium available that could operate at 10 Mbps was coaxial cable.

* Thick wire coaxial cable was installed since this can be used in relatively long lengths of up to 500m before the transmitted / broadcast signal need to be repeated.

* This involves the attenuated signal received at the extremity of the cable segment being amplified and restored to its original form before it is retransmitted out onto the next cable segment.

* Up to five cable segments - & hence four repeaters - can be used in this way. Hence the maximum length of cable the signal propagates is 2.5km plus 4 repeaters which is the origin of the slot-time figure used in the standard.

- 5
- * This disadvantage of thick wire coaxial is that it's relatively difficult to bend and hence install.
 - * To overcome this, thin wire coax was used but, because of the increased (electrical) resistance associated with it, the maximum length of cable for each segment is reduced to 200m.
 - * The arrival of inexpensive adaptive crosstalk canceler circuits to overcome near-end crosstalk (NEXT). It is possible to obtain bit rates of tens of Mbps over twisted-pair cable of up to 100m in length.
 - * It was found that in a vast majority of offices, the maximum length of cable used for telephony to reach each desktop from the wiring closet was less than 100m. Hence unshielded twisted-pair (UTP) cable - as used for telephony - has rapidly become the standard for use with Ethernet. The configuration used for each segment.
 - * Since the cable forms a physical bus, both thick and thin wire coaxial cable installations involve the cable passing near to each attached station. As we can see the figure, with twisted-pair cable a star configuration is used with the hub located in the wiring closet and each station connected to it by means of twisted-pair drop cable. Normally, category three (CAT3) UTP cable is used as for telephony.
 - * Each cable contains four separated twisted pairs. In the case of Ethernet, just two pairs are used: one pair for transmissions from the station to the hub and the second pair for transmission in the reverse direction.
 - * To emulate the broadcast mode of working associated with CSMA/CD the repeater electronics within the hub repeats and broadcasts out the signal received from each of the input pairs onto all of the other output pairs. Hence the signal output by any of the station is received by all the other stations.

As a result, the carrier sense function simply involves the MAC unit within each station determining whether a signal is currently being received on its input pair.

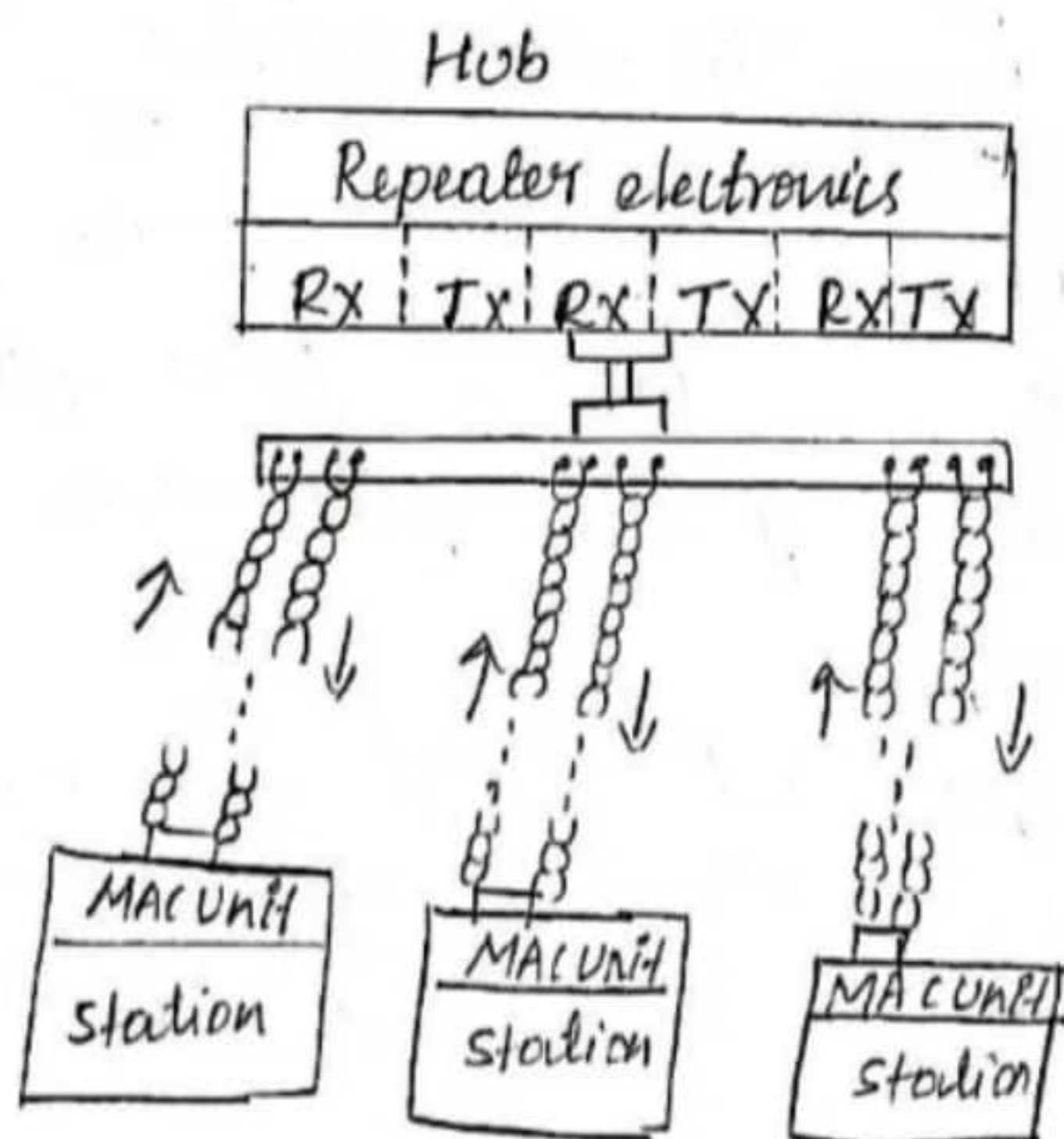
* Similarly, the collision detection function involves the station determining if a signal arrives on its input pair while it is transmitting a frame on the output pair.

* Because of their mode of operation, this type of hub is called a repeater hub and typical numbers of attached stations

* sockets are from 8 through to 16. Above this number multiple hubs are stacked together and are connected by repeaters.

* In the case of repeaters the maximum length of cable between any two stations including the 100m drop cable must not exceed 1.5 km.

a]



b]

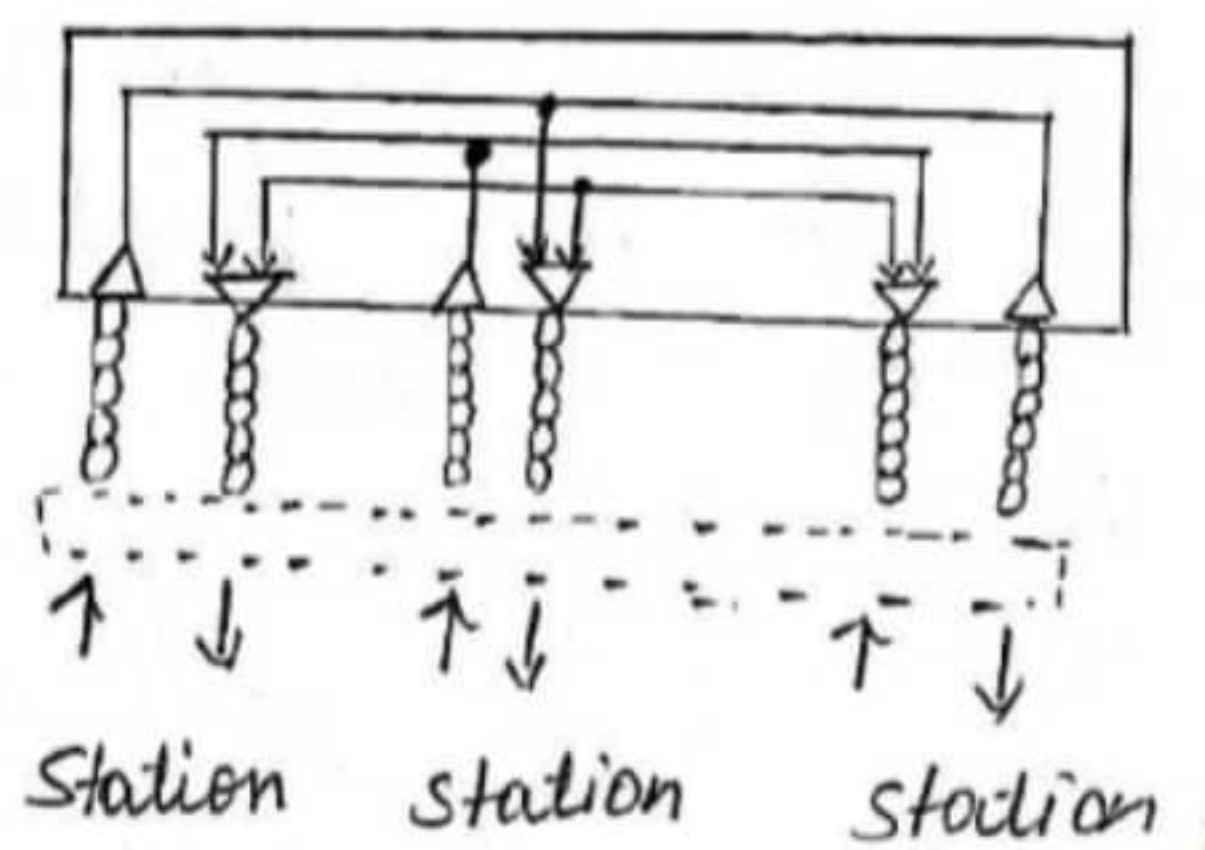
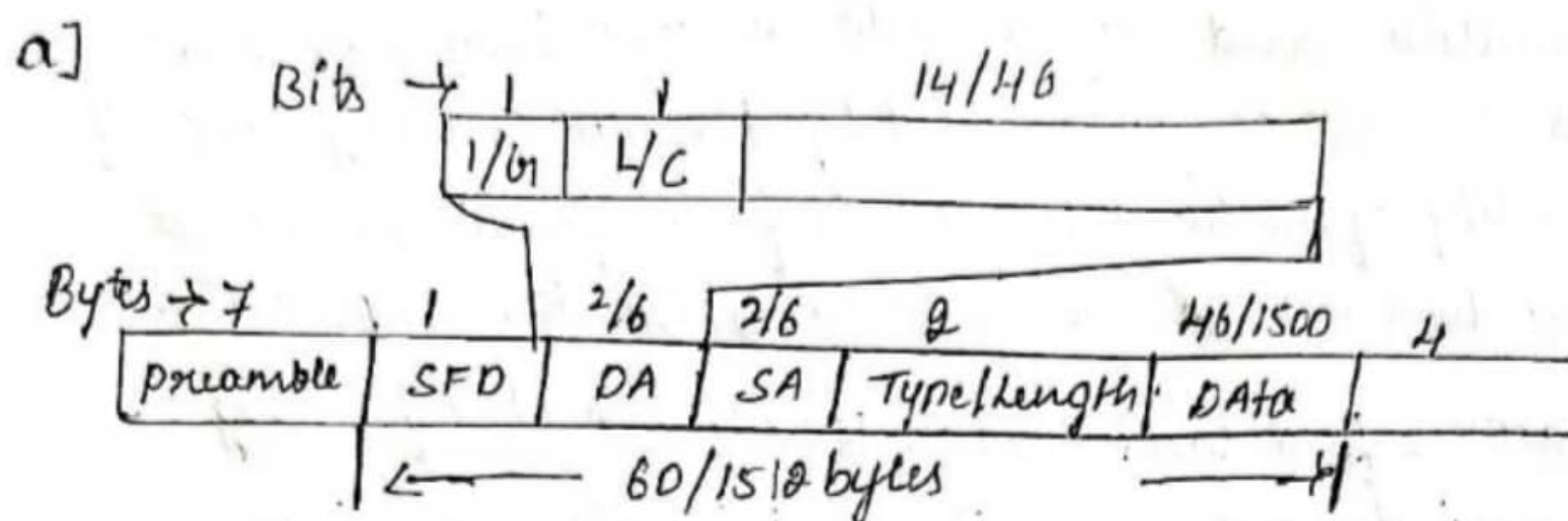


Figure :- Hub configuration principles

a) topology

b) Repeater schematic.



SFD = start frame delimiter

DA/SA = source / destination address

1/b1 = individual [=0] / group [=1] address

FCS = frame check sequence

Hb = locally administrated [=1] / centrally administrated [=0]

b)

Bit rate 10 Mbps [Manchester encoded]

Slot time 512 bit times

Interframe gap 9.6 microseconds

Attempt limit 16

Back off limit 10

Jam size 32 bits

Maximum frame size [including FCS] 1518 bytes

Minimum frame size [including FCS] 512 bits

Figure: Ethernet / IEEE 802.3 characteristic

a) frame format

b) operational parameters

The destination and source address - also known as MAC address because they are used by the MAC Sublayer - specify the identity of the hardware interface of both the intended destination station(s) and the originating station respectively.

- * Each address field can be either 16 or 48 bits, but for any particular LAN installation the size must be the same for all stations.

- * The first bit in the destination address field specifies whether the address is specific an individual address (=0) or a group address (=1). If an individual address is specified the transmitted frame is intended either for a logically related group of stations or for all other stations connected to the network.

- * In the latter case, the address field is set to all binary 1s and for a group address, the address specifies a previously agreed group of stations.

- * The type of grouping is specified in the second bit and can be locally administered (=1) or centrally administered (=0).

- * Group addresses are used for multicasting and the MAC unit/circuit associated with each station in the multicast group is then programmed to read all frames with this group address at its head.

- * If this value is less than the minimum number required for a valid frame, a sequence of bytes is added, known as padding.

- * The maximum size of the data field - normally referred to as the maximum transmission unit (MTU) - is 1500 bytes

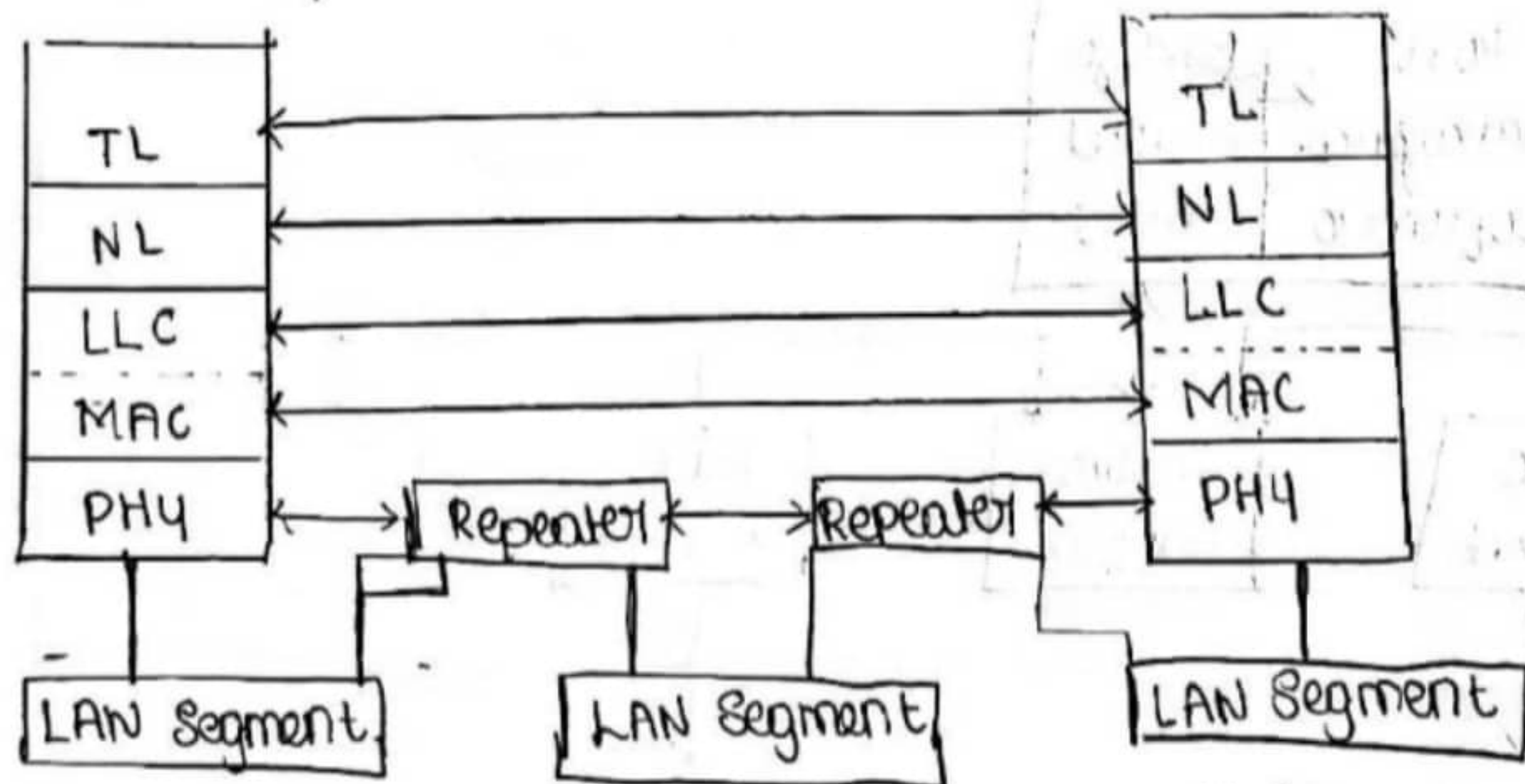
Ranya S

Bridges: The device that can be used to interconnect two separate LANs is known as a bridge.

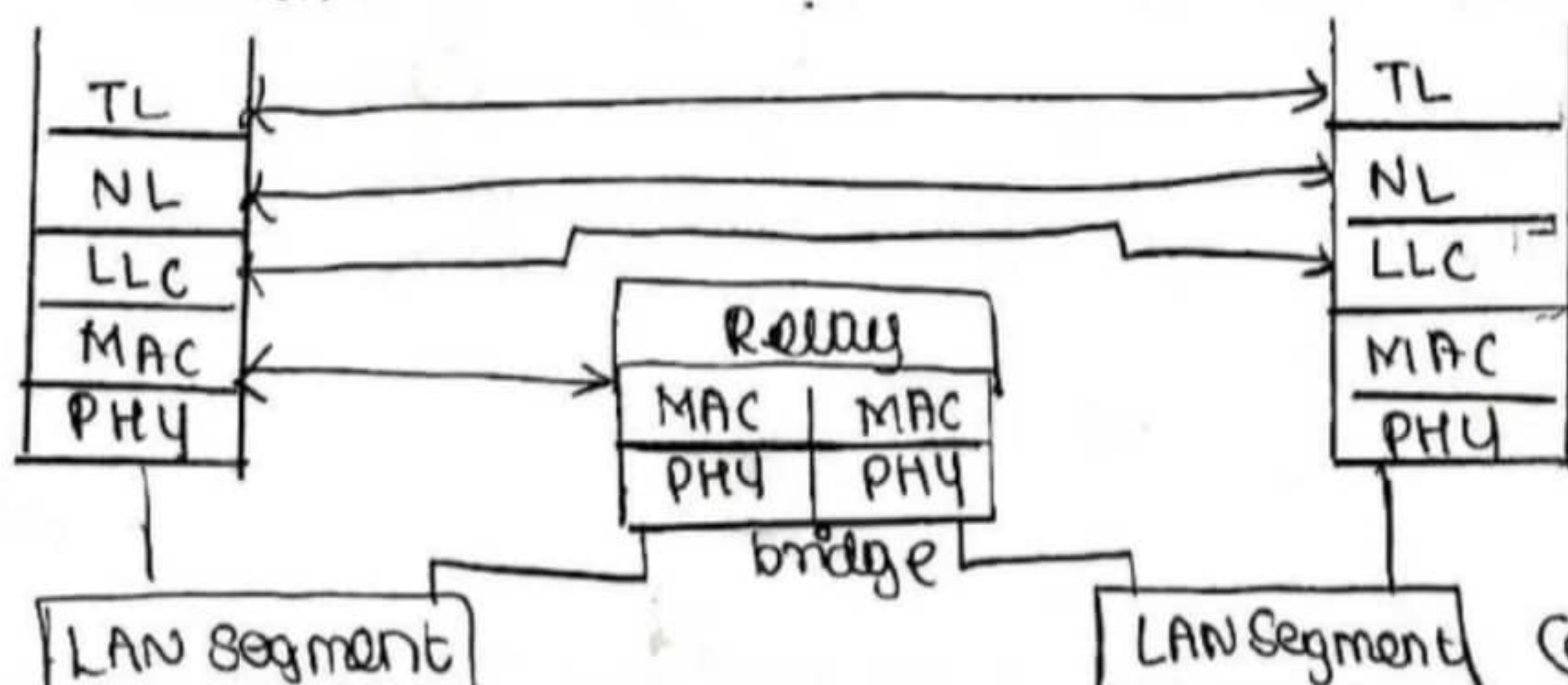
Since bridges were designed to overcome the limitations that occur when using repeaters.

The function of a bridge is similar to the repeaters in that it is used for interconnecting LAN segments. When bridges are used, all frames received from segment are buffered (stored) & error checked before they are repeated (forwarded). Only frames that are free of errors & are addressed to stations on a different segment from the one on which they were received are forwarded. Consequently all transmissions between stations connected to the same LAN segment are not forwarded & hence do not load the rest network.

(a) End Station



(b) End Station



LAN interconnection
fig 10.1
(a) repeaters (b) bridge

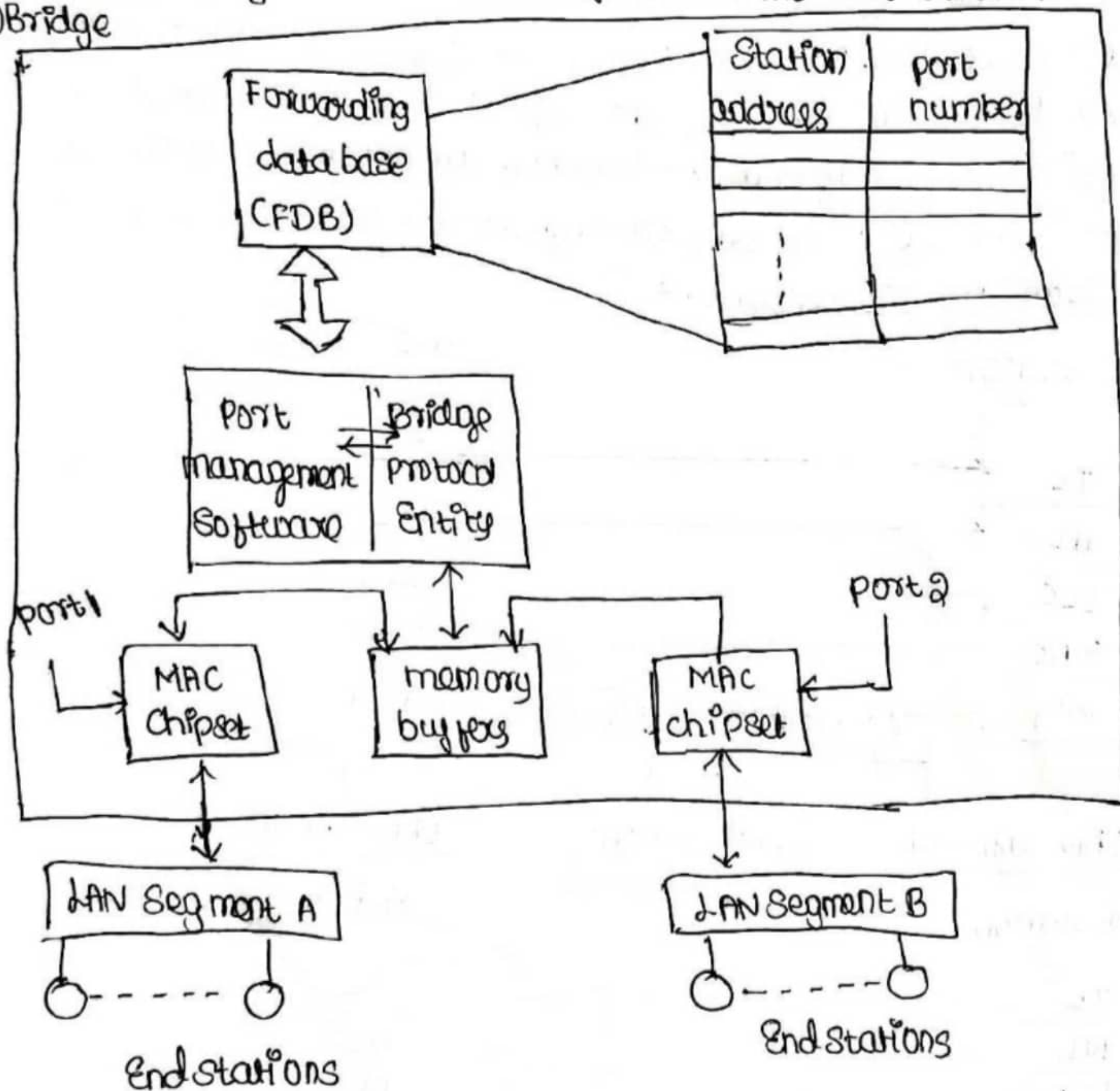
There are two types of bridge the ones that are used for with Ethernet LANs known as transparent bridge.

The others with token ring LANs known as source routing bridge.

Transparent bridge:-

An with a repeater, the presence of one (or more) bridges in a route between two communicating stations is transparent to the two stations.

(a) Bridge



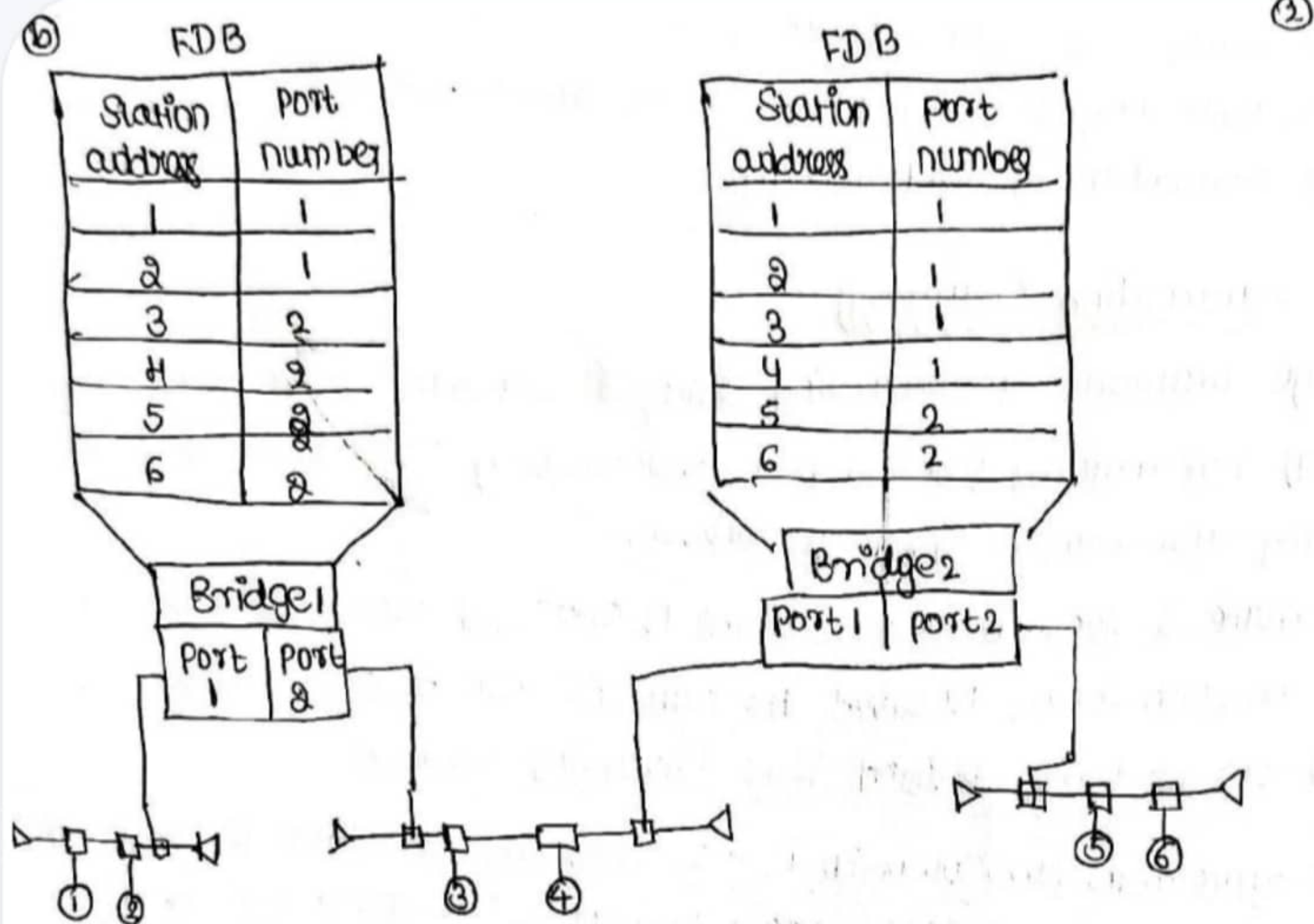


Fig: Transparent bridge Schematic (a) architecture (b) application Example

A LAN Segment is physically connected to a bridge through a bridge port. A basic bridge has just two ports whereas a multiport bridge has number of connected ports.

* Each bridge port comprises the MAC Integrated circuit chipset associated with the particular type of LAN segment - CSMA/CD, Ethernet - together with some associated port management software.

* The software is responsible for initializing the chipset at start up chipsets are all programmable devices for buffer management.

* The available memory is logically divided into a number of fixed size units known as buffers. Buffers management involves passing a free buffer (pointer) to the chipset for onward transmission.

* Every bridge operates in the promiscuous mode which means it receives & buffers all frames received on each of its ports. When a frame has been received at a port & put into the assigned buffer by the MAC

the pool management software prepares the chipset for a new frame & then raises the pointer of the memory buffer containing the received frame to the bridge protocol entity for processing.

Frame Forwarding (Filtering)

A bridge maintains a forwarding data (also known as a routing directory) that indicates for each port, the outgoing port to be used for forwarding each frame received at that port.

If a frame is received at a port that is addressed to a station on the segment on which it was received the frame is discarded; otherwise it is forwarded via the port specified in the forwarding database.

If the same as the port on which it was received, the frame is discarded else it is queued for forward transmission on the segment associated with the accessed port. This process is also known as frame filtering.

Bridge learning

A problem with transparent bridge is the creation of the forwarding database. One approach is for the contents of the forwarding database to be created in advance & held in a fixed memory such as programmable read only memory.

The learning process is as follows. When a bridge first comes into service, its forwarding database is initialized to empty, whenever a frame is received, the source address within it is read & the incoming port number on which the frame was received is entered into the forwarding database.

The incoming port number is entered in the forwarding database against source address & a copy of the frame is forwarded on all the other output ports of the bridge. This action is referred to as flooding since it ensures that a copy of each frame's transmitted is received on all segments in the total LAN.

The learning process works only if the total bridged LAN has a simple (spanning tree) topology. This means that there is only a single path b/w any two segments in the network. ③

Multiple paths b/w two segments cannot exist with the basic learning algorithm. Since the flooding operation during the learning phase would cause entries in the forwarding database to be continuously overwritten.

The simple LAN topology shown figure below.

If Station 10 transmits a frame on Segment 1 during phase learning, then bridge B1 & B2 will both create an entry in their forwarding database - Station 10/Port 1 - & forward a copy of the frame onto Segment 2. Each of these frames will then be received by the other bridge & an entry of these will be made of station 10/Port 2 & a copy of the frame output at Port 1. Each of these will be received by the other bridge resulting in their corresponding entry for Port 1 being updated. The frame will thus continuously circulate in a loop with entries for each port being continuously updated.

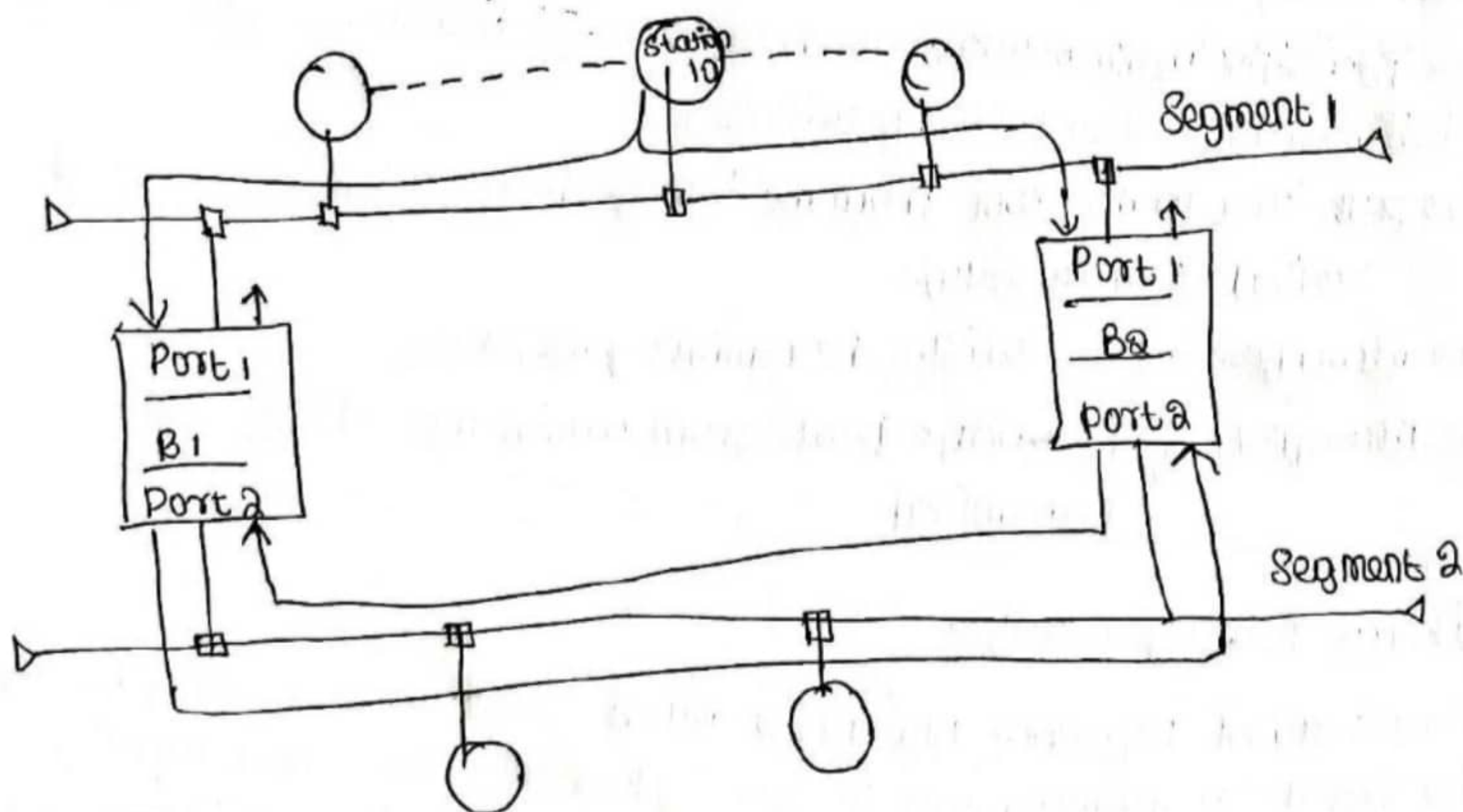


fig: Effect of dual paths on learning algorithm.

The resulting logical or active topology behaves as a single spanning tree & the algorithm is known as spanning tree algorithm.

Spanning tree algorithm

With the spanning tree algorithm all the bridges regularly exchange special frames (messages) known as bridge protocol data units (BPDUs). Each bridge has a priority value & unique identifier. For the total bridged LAN, a single bridge is dynamically chosen by the spanning tree algorithm to be the root bridge. This is the bridge with the highest priority & the smallest identifier. It is determined / confirmed at regular intervals.

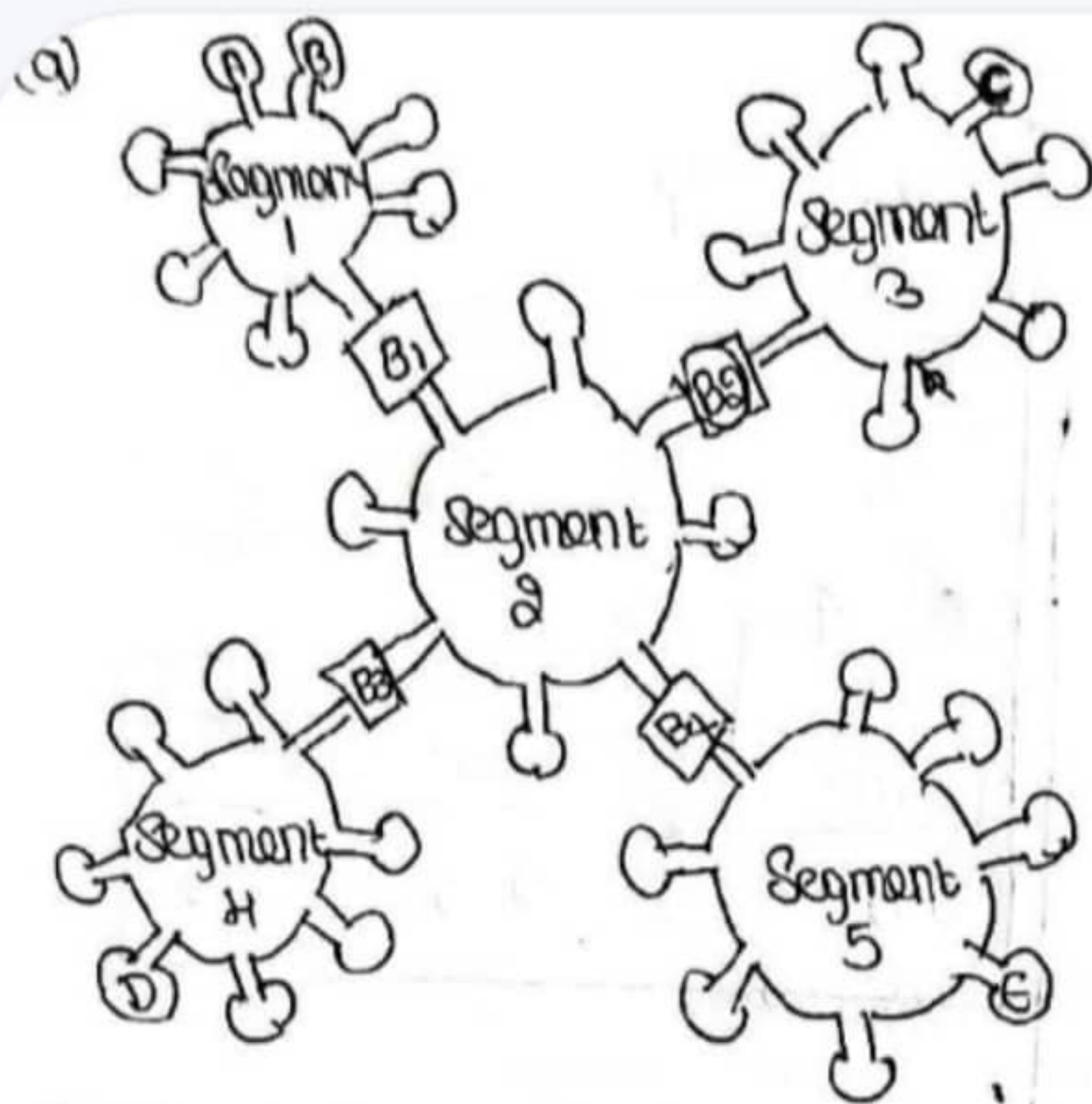
When a bridge is first brought into service, it assumes it is the root bridge. A bridge that believes it is the root (all initially) initiates the transmission of a configuration BPDU on all of its ports at regular time intervals known as the hello time.

Each configuration BPDU contains a number of fields including

- the identifier of the bridge which the bridge transmitting the BPDU believes to be the root (itself initially).
- * the path cost to the root from the bridge port on which BPDU was received (zero initially).
- * the identifier of the bridge transmitting the BPDU
- * the identifier of the bridge port from which the BPDU was transmitted.

Source routing bridges

The major difference b/w a LAN based on source routing bridges & one based on spanning tree bridges is that with the latter bridges collectively perform the routing operation in a way that is transparent to the end station.



(b) Routing table held by A: Destination B = 8 segments

$C = \text{Segment 1, B1, Segment 2, B2, Seg 3}$

$E = \text{Segment1, B1, Segment2, B4, Seg5}$

Routing table told by B: Best'nation A = Segment

$\mathcal{D} = \text{Segment 1, B1, Segment 2, B3, Segment 4}$

Routing table held by E: Destination A = Segment 5,
B, Segment B; Seg.

C = Segment 8, B4, Segment 1
B2, Segment 6

Fig: An Example source routing bridged LAN

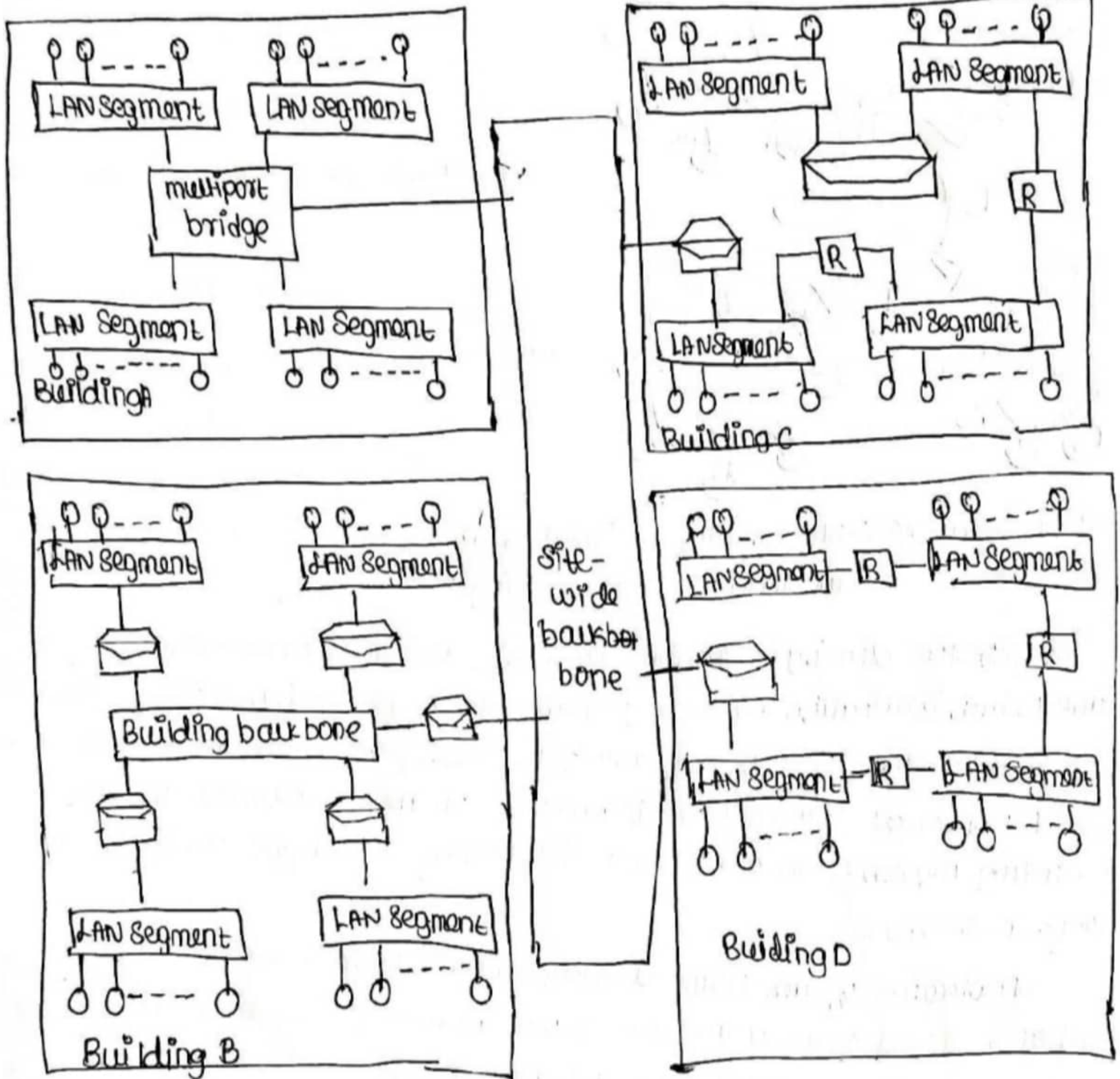
(a) topology

(b) Flowing table Entries

With source routing, a station ascertains the route to be followed by a frame to each destination before any frame are transmitted. This information is inserted at the head of the frame & is used by each bridge to determine whether a received frame is to be forwarded on another segment or not. The routing information comprises a sequence of segment-bridge segment-bridge identifiers.

On receipt of each frame, a bridge needs only to search the routing field at the head of the frame for its own identifier. The frame is repeated at the ring interface by the bridge if forwarded. The address recognized & frame copied (c) bits in the frame status (FS) field at the tail of the frame are set to indicate to the source station (bridge) that it has been received (forwarded) by the destination station (bridge).

FDDI

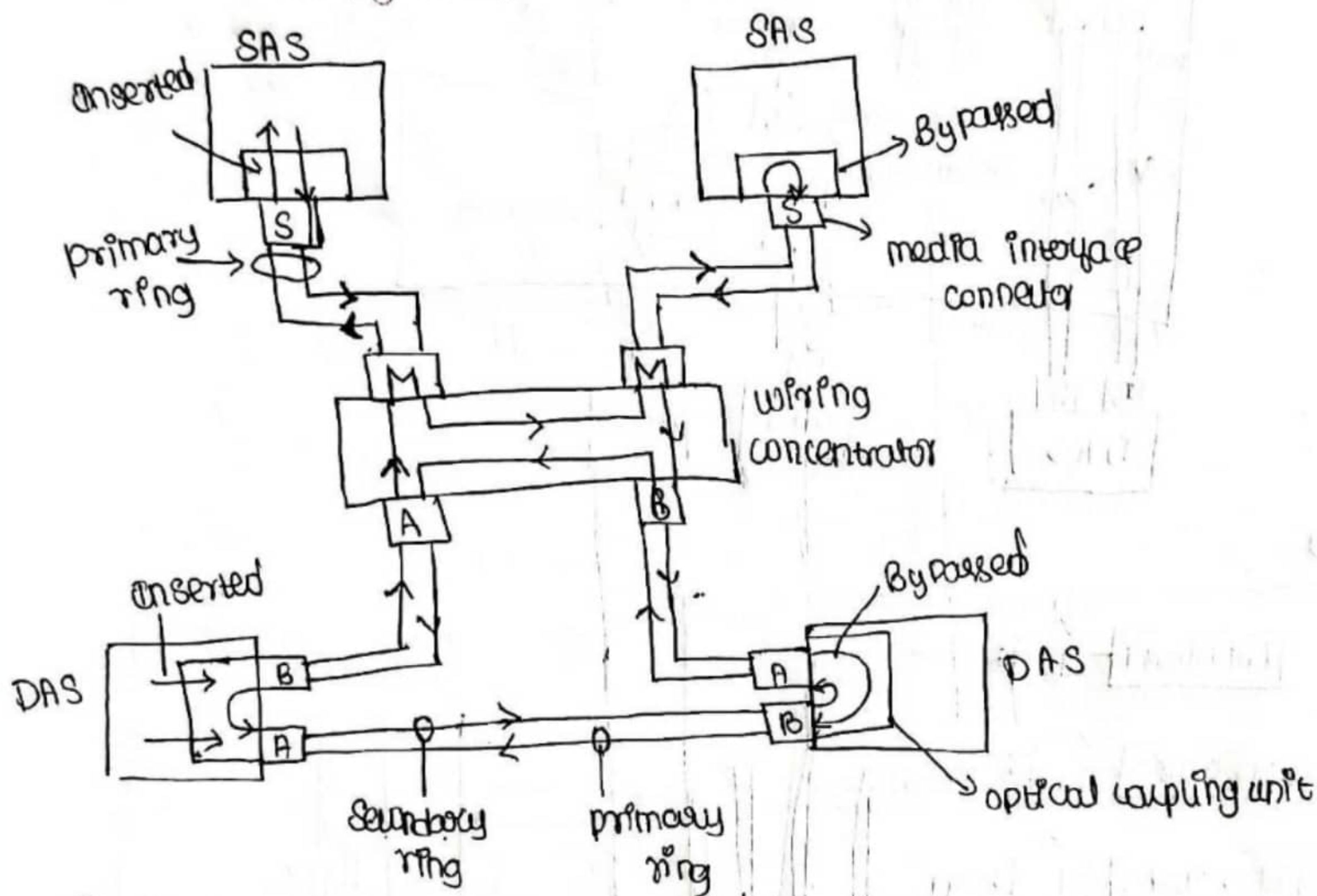


○ = End Station (workstations, servers, etc). (R) = repeater ☐ = Bridge

FDDI: Fiber Distributed Data Interface (FDDI) is a token passing ring network that operates at 100 mbps on optical fiber-medium. It can be used for the interconnection of segments spread over a wider geographical area than a single building. The resulting h/w is known as an establishment or site backbone.

The FDDI LAN Std was developed by the American national standards Institute (ANSI). It is based on a ring topology & operates at a bit rate of 100Mbps. Eg: digitized speech & video.

Network configuration:



SAS: Single attach Station

M: master key

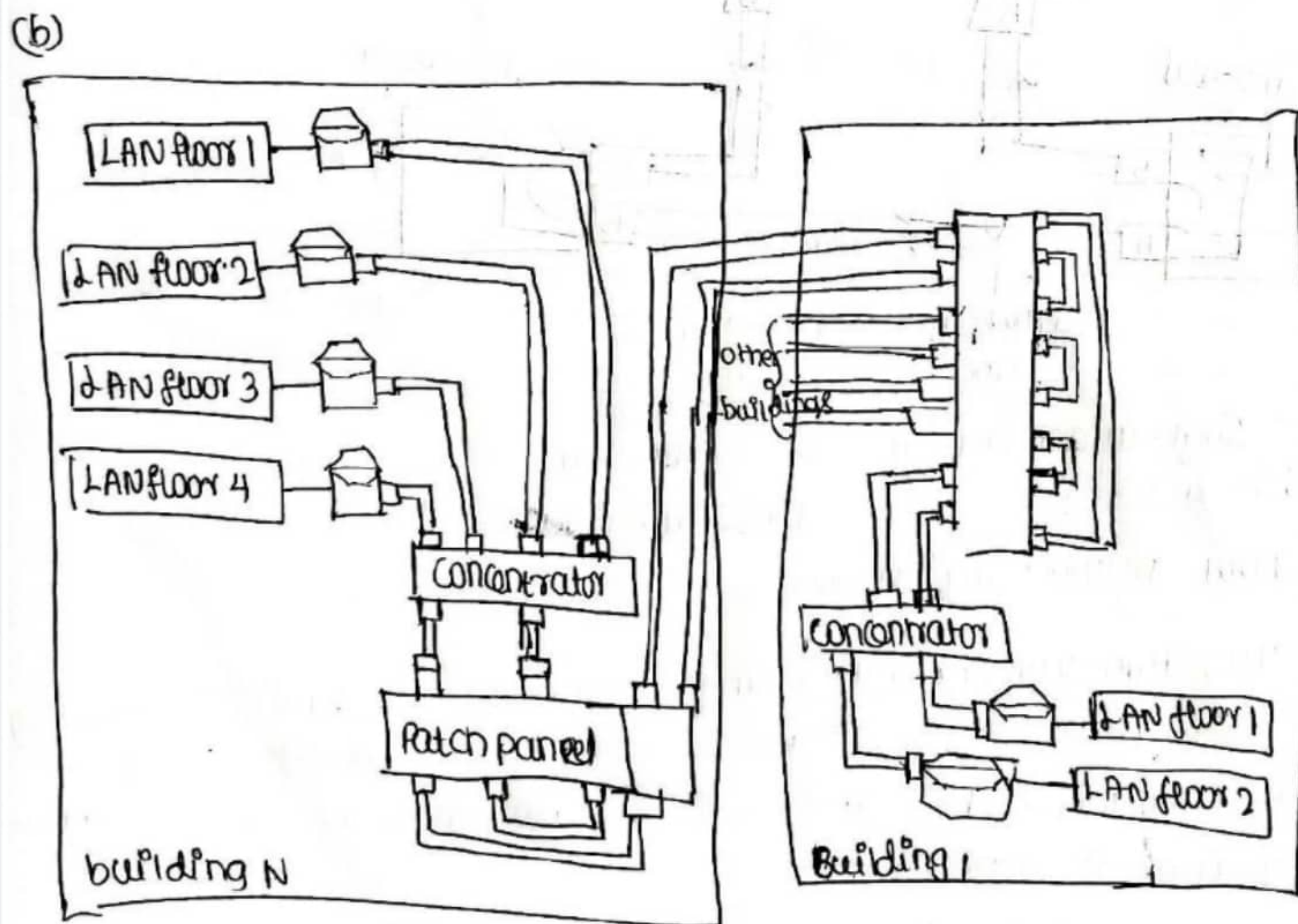
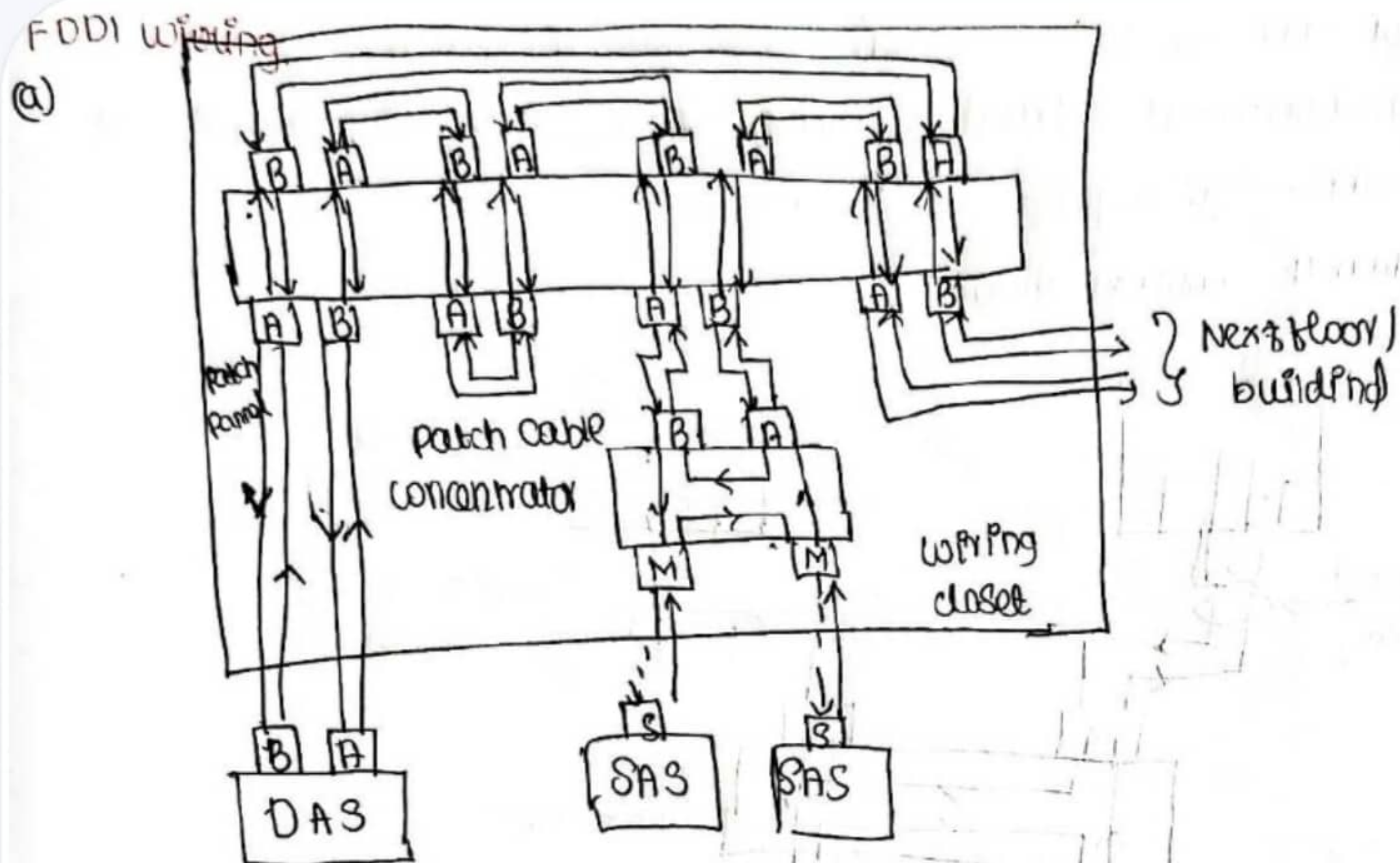
S = slave key

DAS: dual attach Station

Fig: FDDI networking components.

FDDI uses two counter rotating rings to enhance reliability: the primary ring & the secondary ring. The secondary ring can be used either as an additional transmission path or purely as a backup in the event of a break occurring in the primary ring.

There are two types of stations: a dual attach station (DAS) which is connected to both rings & a single attach station (SAS) which is attached only to the primary ring. Most user stations are attached to the ring via wiring concentrators since then only a single pair of fiber is needed & connection cost is lower.



Bridge

fig: FDDI wiring schematic (a) building (b) establishment

Although the topology is logically a ring, physically it is normally implemented in the form of a hub tree structure. To ensure that changes to the wiring are carried out in a controlled way, a combination of patch panels & wiring concentrators are used.

Each patch panel has a number of possible attachment points associated with it. In the absence of a connection at a particular point the ring is maintained using short patch cables each with the same type of connectors. Adding a new station or concentrator simply involves removing a patch cable & replacing it with a corresponding drop cable. This approach is known as structured wiring.

Physical interface

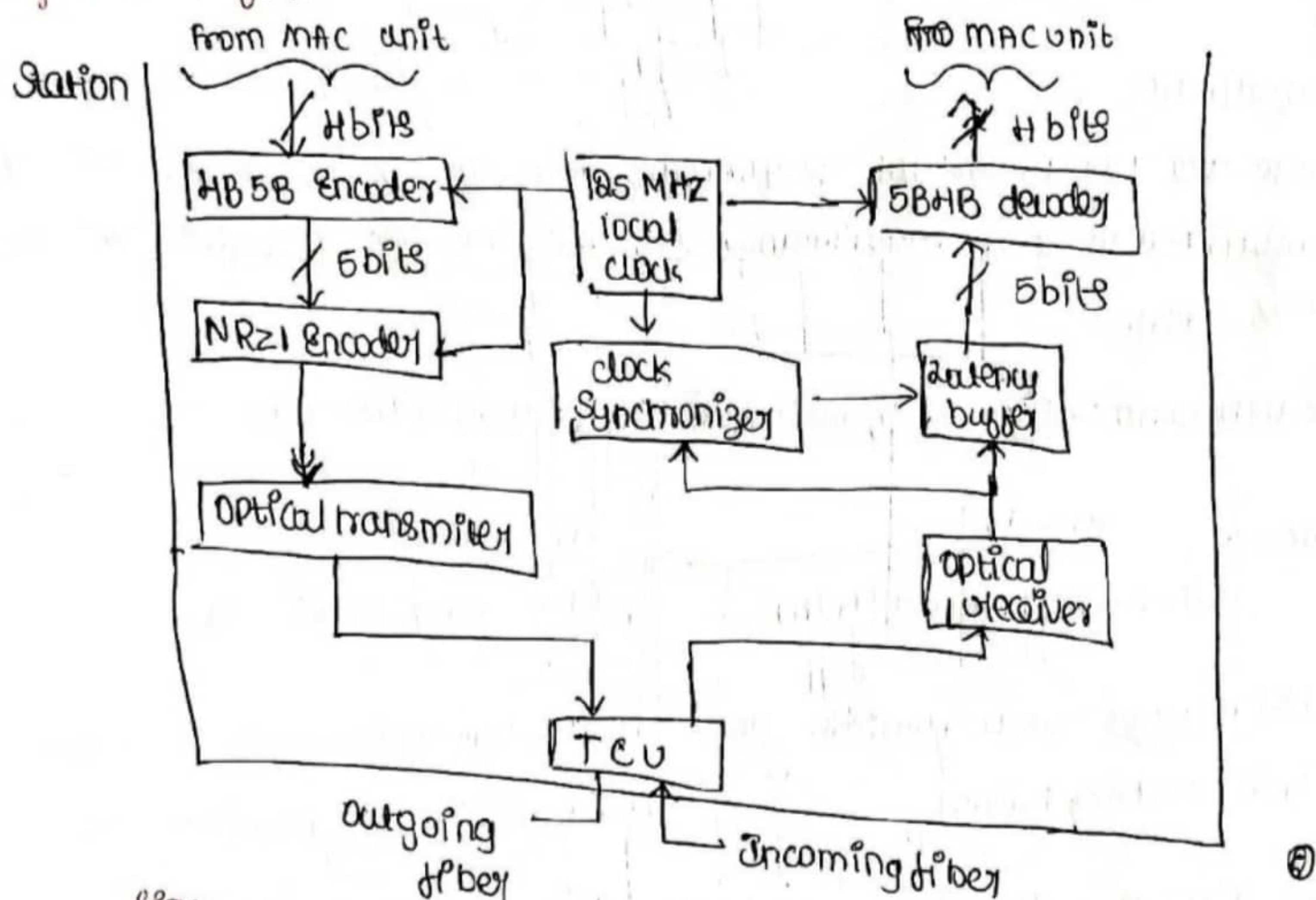
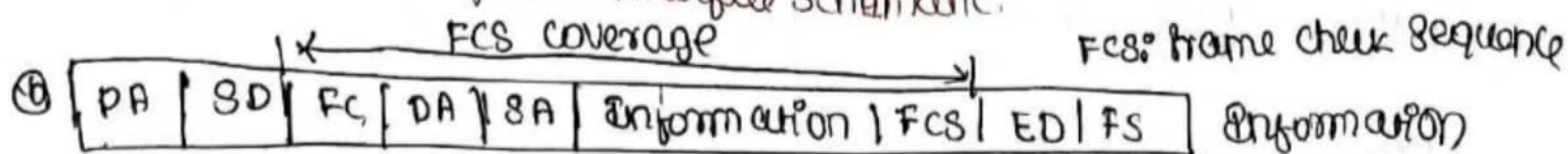


Fig: FDDI physical interface schematic.



PA | SD | FC | ED | TOken

SA: Source address

Fig: frame format

PA: Preamble (16 or more symbols)
SD: Start delimiter (2 symbols)
FC: frame control
DA: Destination address

The preamble (PA) field consists of 16 or more idle symbols which since each consist of five 1 bits, cause the line signal to change at the maximum frequency. 1

The Start delimiter (SD) field consists of two control symbols which enable the receiver to interpret the following frame contents on the correct symbol boundaries. The FC, DA & SA fields have the same meaning as before but the information field in the data frames can be up to 1500 octets with FDD. The End delimiter (ED) field contains one or two control symbols. The frame status (FS) field consists of three symbols that are combination of the two control symbol R & S.

Performance

There are two important performance measures associated with shared medium networks: the maximum obtainable throughput & the maximum access delay.

The maximum utilization of the nominal output capacity U_{max} is

$$U_{max} = \frac{T_{TRT} - T_1}{(T_{TRT} - T_1) + T_1 + T_b + (T_1/n)} = \frac{n(T_{TRT} - T_1)}{n(T_{TRT} - T_1) + (n+1)T_1 + T_b}$$

T_{TRT} : Target token rotation time

T_b : The time to transmit the token

T_1 : the ring latency

n : number of station in the ring

The T_{TRT} is very much greater than T_b , $\therefore$ the maximum utilization can be given as

$$U_{max} = \frac{n(T_{TRT} - T_1)}{nT_{TRT} - T_1} \quad \text{or approximately} \quad \frac{T_{TRT} - T_1}{T_{TRT}}$$

the expression to compute the maximum access delay of an FDDI ring

$$A_{max} = (n-1) (T_T R_T - T_i) + nT_i + T_i \\ = (n-1) T_T R_T + 2T_i$$

where A_{max} is the worst-case time to receive a usable token.

Timed token rotation protocol

An FDDI ring uses a scheme that is based on a timed token rotation protocol that is controlled by preset parameters known as target token rotation time (TTRT).

High-Speed LANs

As the application of LANs has become more diverse, so the demands on them in terms of information/data throughput have increased. As we have just described by using a combination of bridges & a high bit rate backbone, the throughput of the LAN is determined by the maximum throughput of each LAN segment.

A number of variations of Ethernet are

- Fast Ethernet
- Switched Fast Ethernet
- Gigabit Ethernet

Fast Ethernet:

The aim of fast Ethernet was to use the same shared, half duplex transmission mode as Ethernet but to obtain a 10 increases in operational bit over 10 base T while at the same time retaining the same wiring system & MAC method & frame format.

Line code

The major technological hurdle to overcome with fast Ethernet was to achieve a bit rate of 100Mbps over 100m of UTP cable.

category 3 UTP cable is used on each pair and four pairs are used to achieve the required bit rate 100Mbps in each direction. Hence it standard also known as 100BaseT.

With the CSMA/CD access control method, in absence of contention for the medium, all transmission are half duplex, that either station to hub or hub-to-station. In a 10BaseT installation just two of the four wire pairs are used for data transfers one in each direction. collision are detected when the transmitting station (or hub) detects a signal on the receive pair while it is transmitting on the transmit pair. Since the collision detect function must also be performed in 100BaseT the same two pairs are used for this function the remaining two pairs are operated in a bidirectional mode.

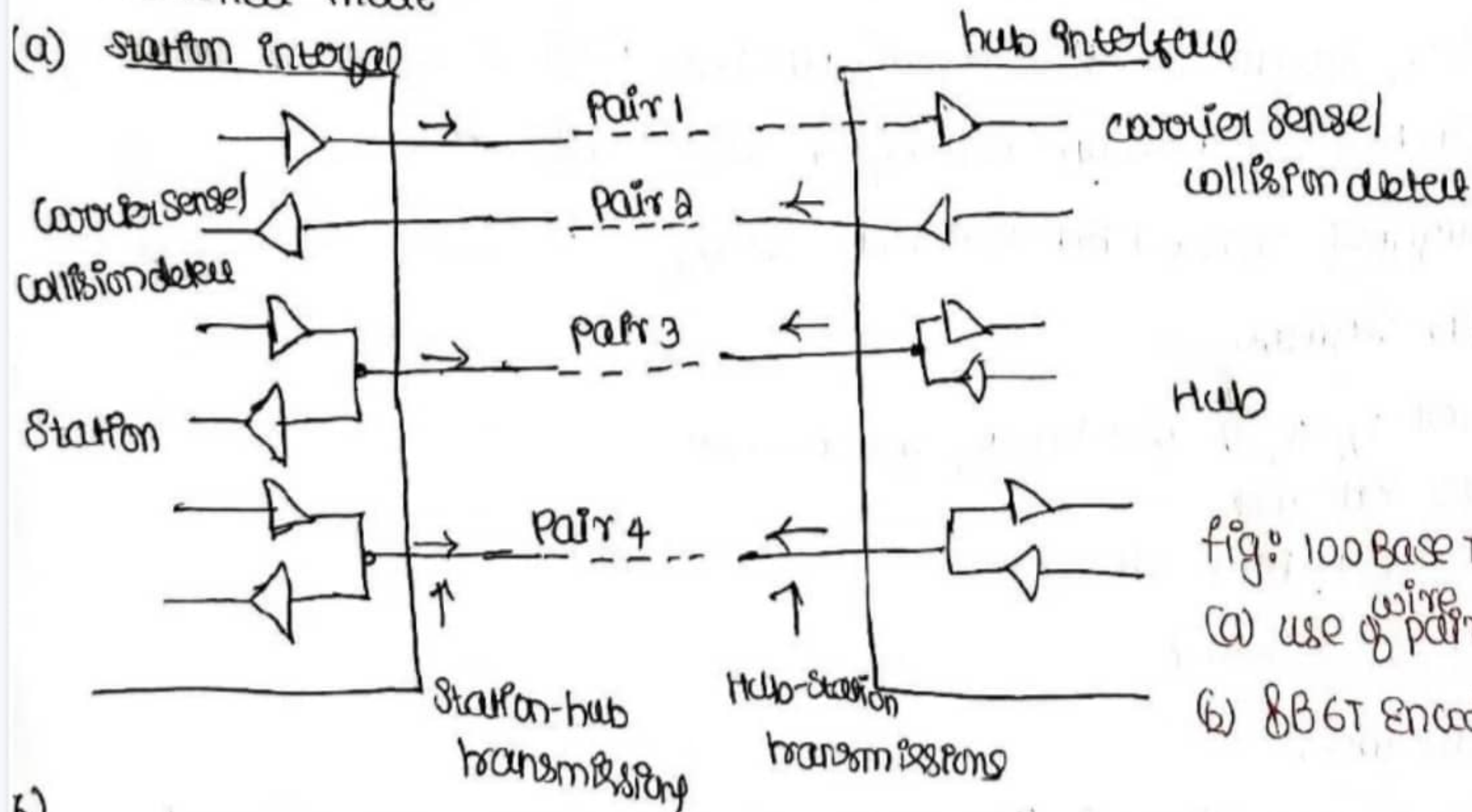
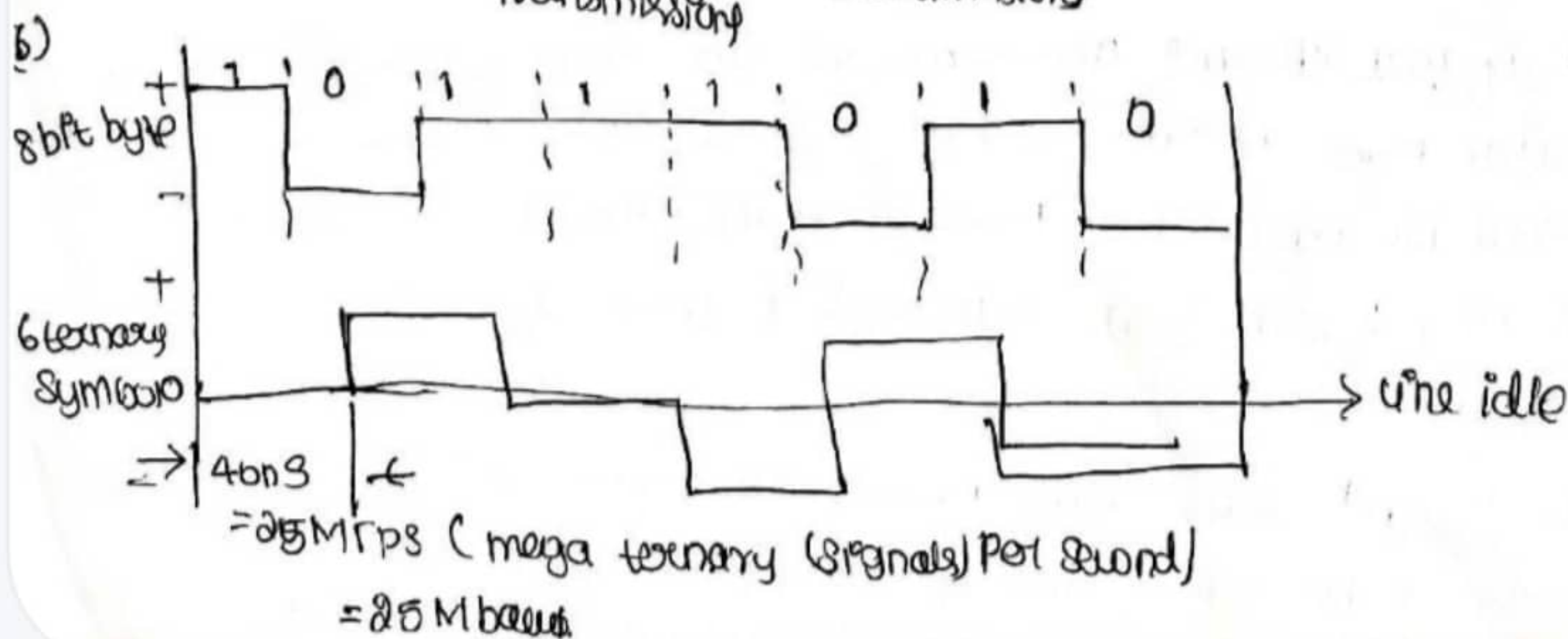


fig: 100BaseT:
(a) use of wire pairs
(b) 8B6T Encoding



If we use Manchester encoding a bit rate of 33.33 Mbps requires a baud rate of 33.33 Mbaud which exceeds the 30 Mbaud limit set for use with such cable as above this unacceptably high levels of cross talk are obtained.

To reduce the baud rate, a 3-level (ternary) code is used instead of straight (2-level) binary coding. The code used is known as 8B6T, which means that prior to transmission each set of 8 binary is first converted into 6 ternary symbols.

DC Balance

As we have just indicated all the code words selected have a combined weight of either 0 or +1. For example the code word $+ - - + 0 0$ has a combined weight of 0 while the code word $0 + + + - -$ has a weight of +1.

If a string of code words each of weight +1 is transmitted then the mean signal level at the receiver will move away rapidly from the zero level causing the signal to be misinterpreted this is known as DC wander & is caused by the use of transformers at each end of the line. The presence of transformers means there is no path for direct current (DC)

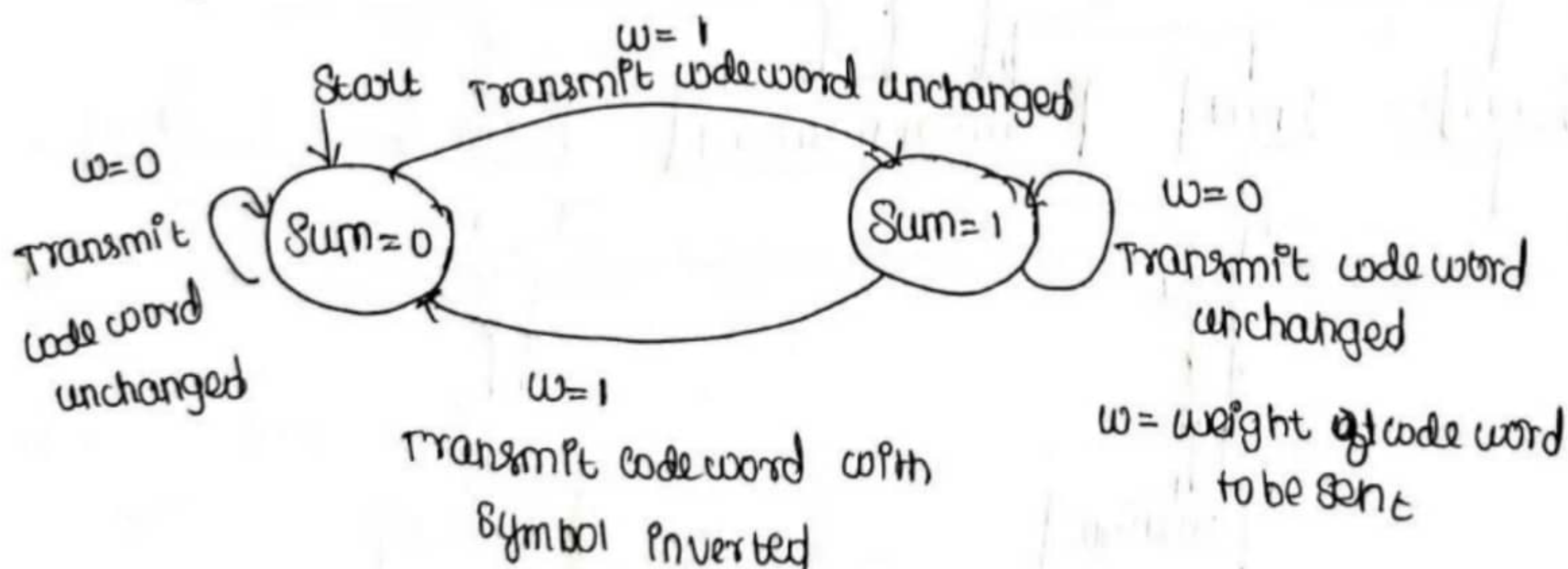


Fig: 100Base-T transmission detail: DC balance transmission rules.

End of frame Sequence

The transmission procedure adopted enables further error checking to be added to the basic CRC at the end of each frame transmission - that is after the four CRC bytes have been transmitted - one of two different end-of-stream (EoS) codes is transmitted on each of the three pairs

Switched Fast Ethernet.

In order to allow multiple access/transfers to be in progress concurrently two developments have been made: the first the introduction of a switched hub architecture & the second duplex working over the cts that connect the stations to the hub. The resulting type of hub is known as a Fast Ethernet Switch.

Switch Architecture

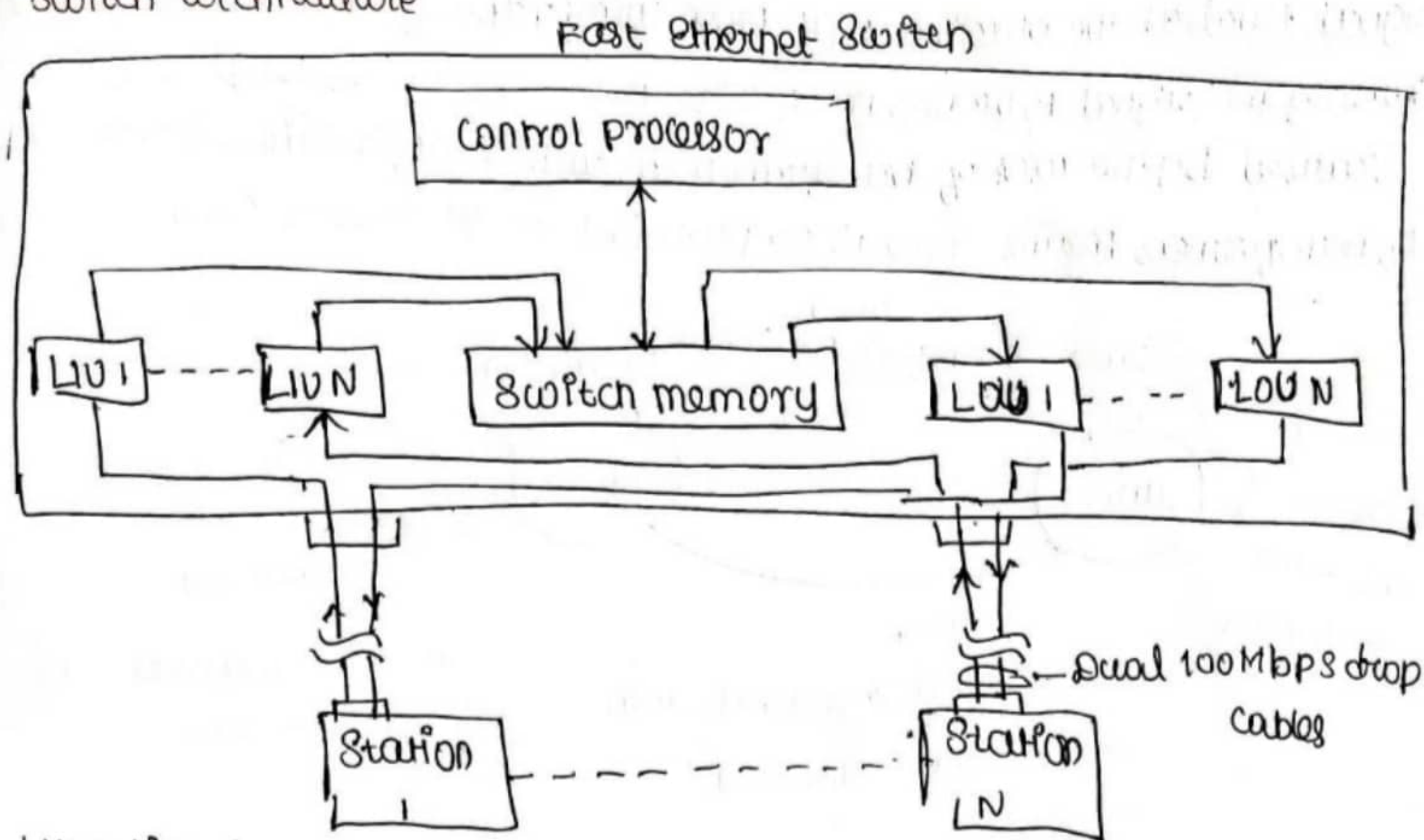


fig: Fast Ethernet Switch Schematic

Each station is connected to the hub by means of a pair of lines which typically are implemented as dual UTP (or STP) cables or dual multimode fibre

in the case ^{case} of dual multimode fiber cable each fiber is used to transmit at 100Mbps over several kilometers one in each direction of transmission. An active signal is maintained on each fiber continuously by transmitting an idle symbol during the idle period b/w frames.

Flow Control

When using flow control should the control processor find that the level of memory in use reaches a defined threshold it initiates the transmission of a pause frame on all of its input ports.

On receipt of a pause frame the attached station must then stop sending any further frames to the switch until either a defined time has expired or it receives a notification from the switch that the overload condition has passed. Having sent a pause frame the control processor monitors the level of memory in use & when this falls below a second level, it sends out a continue frame on all input ports to inform the attached stations that they can now resume sending new frames.

Network Configuration

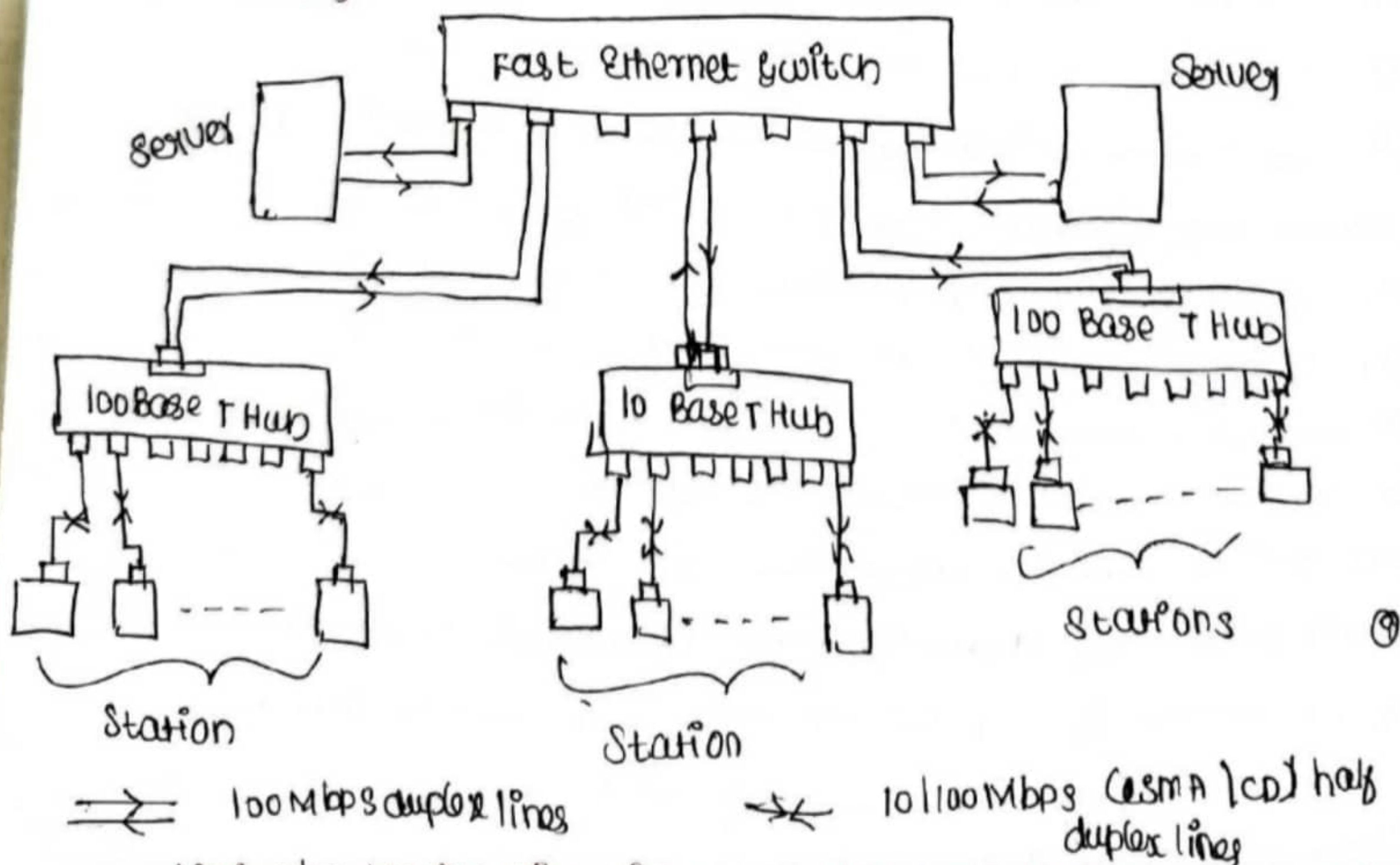


Fig: New configuration with a fast Ethernet switch & independent hub

In order to obtain a high level of throughput, the two servers are connected directly to the switch by means of duplex 100Mbps lines. All the end user stations then gain access to the servers through either a 10BaseT or a 100BaseT hub. Hence the duplex uplink port connecting each hub to the switch is through a bridging hub since this has bridging circuitry within the hub to temporarily buffer all frame transfer to & from the switch & to perform the CSMA/CD MAC protocol associated with the shared medium hub ports. The switch also automatically configures the operating speed of each of its ports to be either 10 or 100Mbps.

Giga bit Ethernet:

As the name implies the drop cable associated with Gigabit Ethernet hubs operate at 1000 Mbps (1Gbps). The standard is defined in IEEE 802.3z & has been introduced to meet the throughput demands of an increasing number of servers that hold files containing multimedia information; eg. include web pages comprising very high resolution graphics, motion video & general audio. The hub can be either a simple repeater hub - that is one that has no memory associated with it - or a switched hub.

As eg. application of a repeater hub is to distribute the output of a powerful super computer to a localized set of work stations. The main issue when operating in the repeater mode is to ensure that the round trip delay between any two stations - the slot time - exceeds the time required to transmit the smallest allowable frame of 512 bits. The time to transmit a 512 bit frame at 1Gbps is 0.512 μ secs. whereas the slot time is order of 2 μ secs. To overcome this a technique known as carrier extension is used. This as the name implies ensure that a known signal is present on the line for the duration of the slot time & $\therefore$ a collision can still be detected. Alternatively a technique called frame bursting can be used which allows multiple short frames to be transmitted ~~to be~~ after the other up to the slot time.

The Internet

- Interconnection of computers and computer networks using TCP/IP communication protocol.
- A protocol is a set of rules defining communication between system

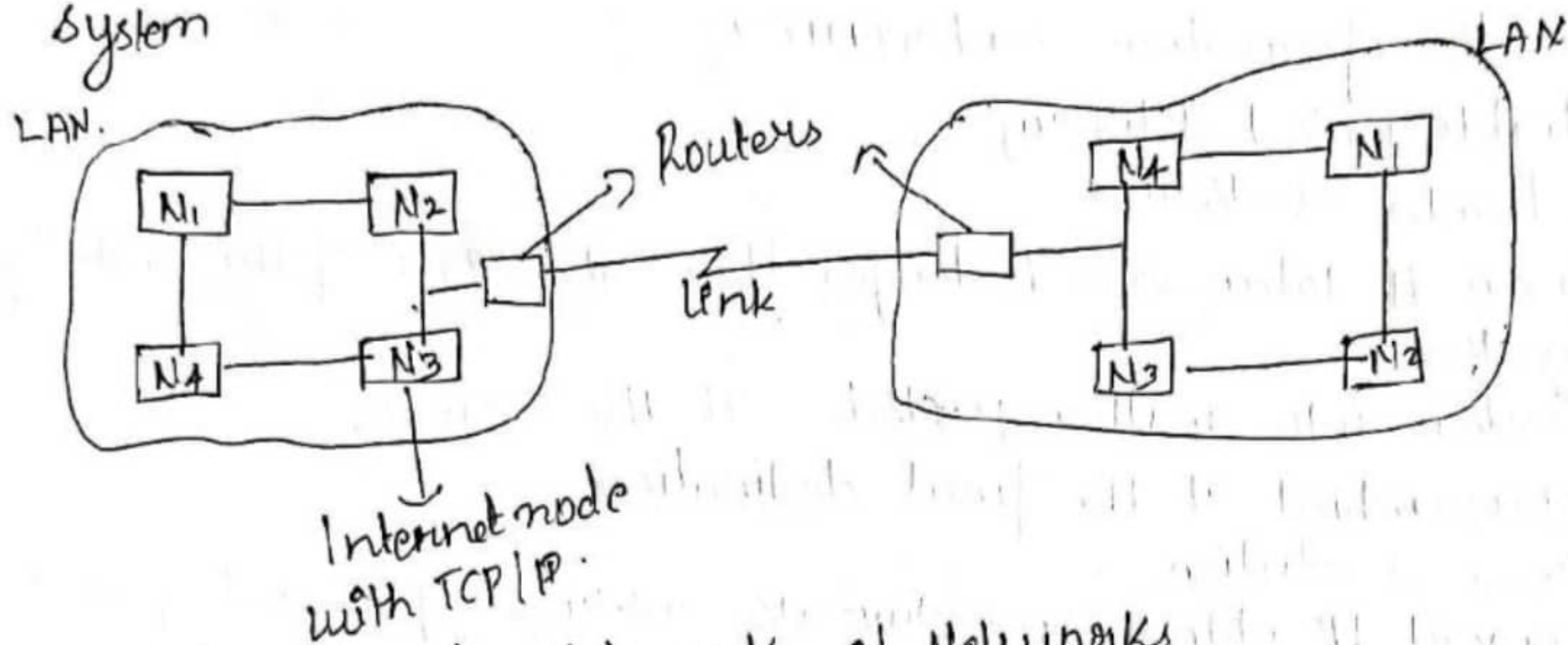


fig : The Internet - Network of Networks

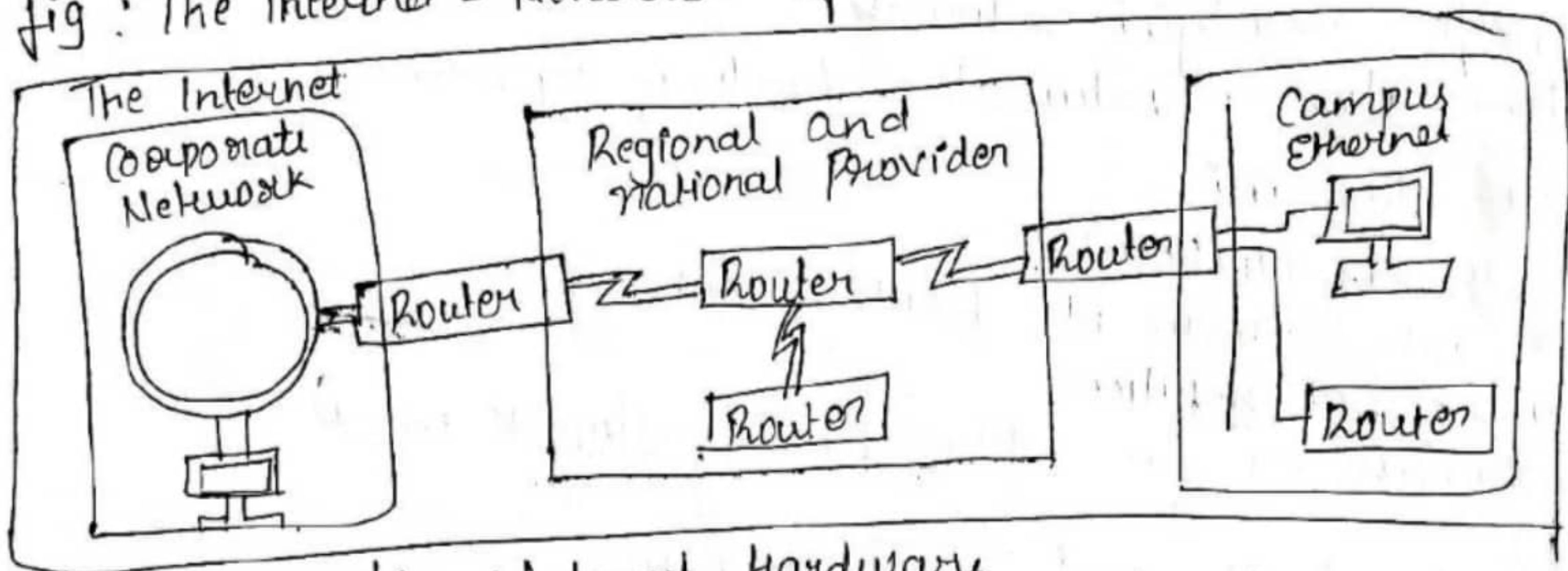


fig : Internet Hardware

Internet Services

- * E-mail
 - ↳ one to one
 - ↳ one to many
- * Telnet - remote login
- * FTP - file transfer
- * Web [HTTP]: Hypertext linking / Navigation
- * IRC: Internet Relay chat
- * Internet telephony, mobile access etc

LAN protocols

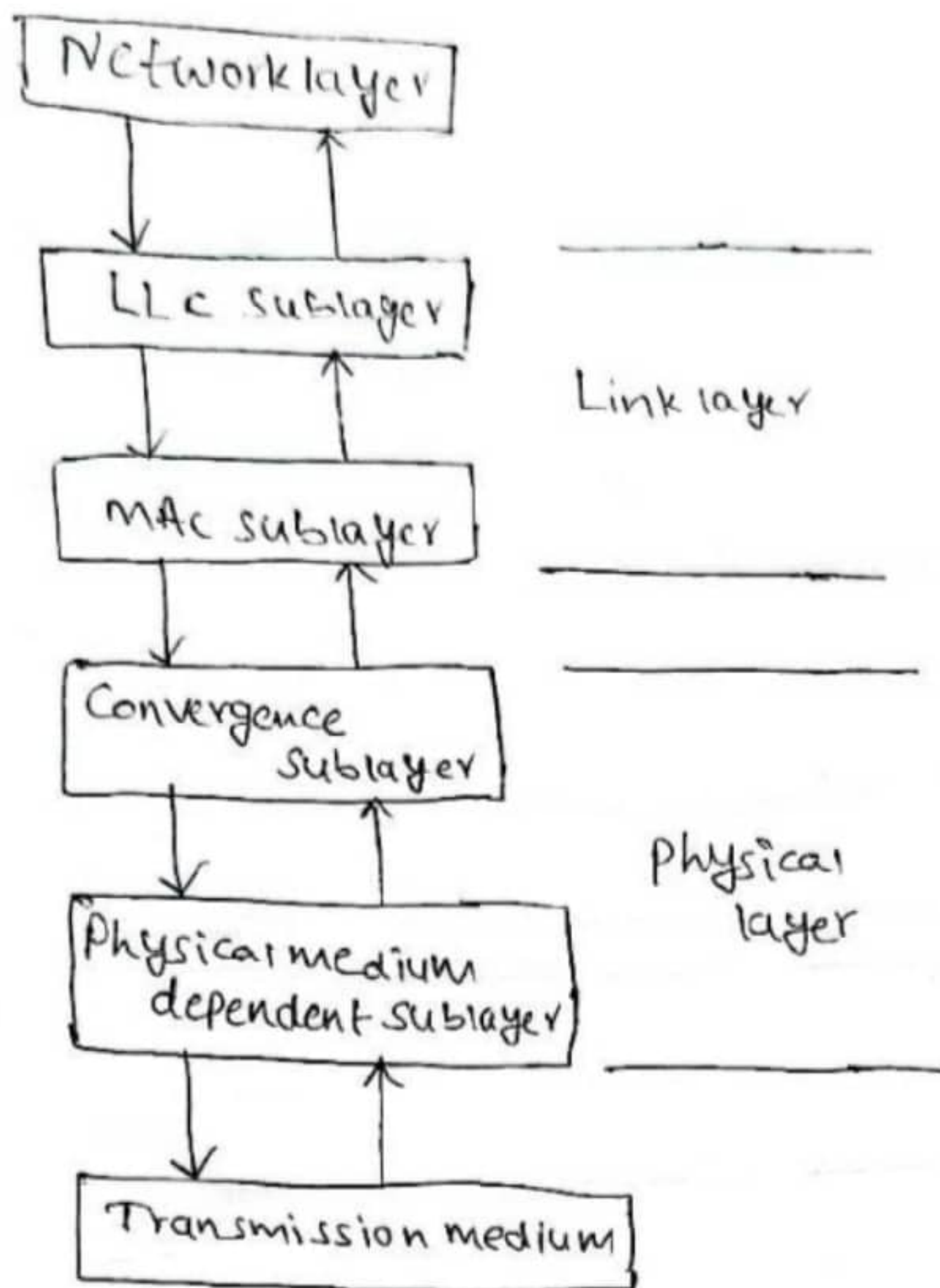


Fig: protocol framework

IEEE

- 802.1 station management
- 802.1d transparent bridges
- 802.2 Logical link control (LLC)
- 802.3 CSMA/CD (Ethernet) bus
- 802.3u Fast Ethernet
- 802.3x Hop-by-hop switch flow control
- 802.3z Gigabit Ethernet
- 802.5 Token ring

Fig: Examples

- Different types of LAN each of which uses a different MAC method, transmission mode and transmission medium.
- The link control sublayer - known as the logical link control (LLC) sublayer in the context of LANs - then offers a standard link layer service to the network layer above it.
- Apart from FDDI, which is defined in ISO9314, the various protocol standards associated with LANs are all part of the IEEE 802 series.

Physical layer

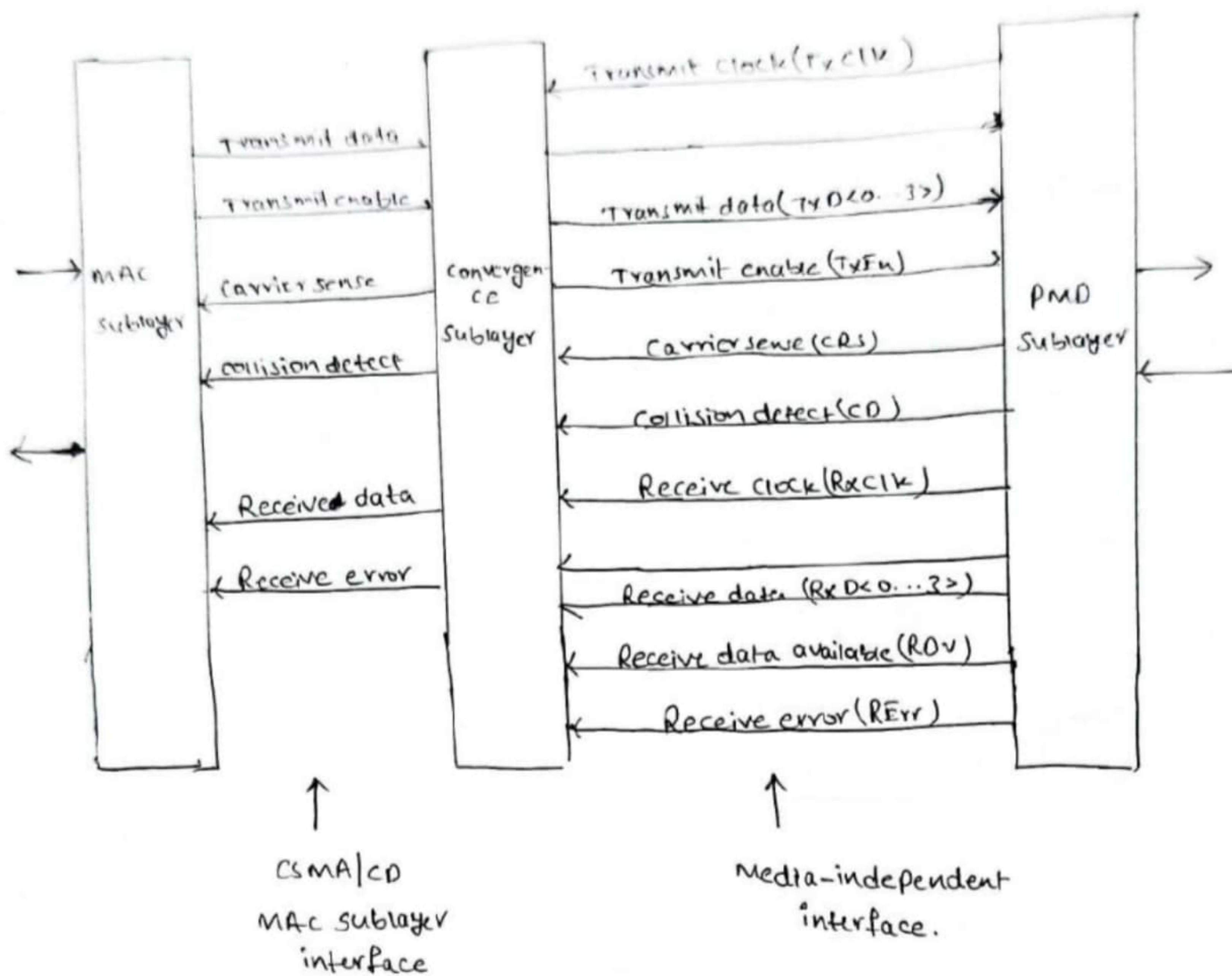


Fig: Fast Ethernet media-independent interface.

→ The Physical layer divided into two sublayers:

- 1) The Physical medium dependent (PMD) sublayer
- 2) (Physical) convergence sublayer (CS).

→ To facilitate use of different media types, a media-independent interface (MMI) has been defined for use b/w convergence and PMD sublayers.

- The set of Signals associated with both interfaces vary slightly for the different types of Ethernet (802.3) and token ring (802.5)
- For example, one 4-bit group when 4B5B coding is used (100Base-SX), and two 4-bit groups when 8B/10B coding is used (1000Base-SX). Hence, all transfers over the MII are in 4-bit nibbles.

MAC Sublayer

Standard set of User Service Primitives defined for use by LLC Sublayer. These are:

MA-UNITDATA.request
MA-UNITDATA.indication
MA-UNITDATA.confirm

- For a CSMA/CD LAN, the ~~com~~ confirm primitive indicates that the block of data associated with the request has been Successfully (or not) transmitted.
- The MA-UNITDATA.request Primitive includes: the required destination address, a service data unit and the required Class of service associated with PDU.
- The MA-UNITDATA.confirm Primitive includes a Parameter that specifies the Success or Failure of associated MA-UNITDATA.request Primitive.
- However in Fig 8.33, the confirm primitive is not generated as a result of a response from the remote LLC sublayer but rather by the local ~~MAC~~ MAC entity.

← Source MT

LLC sublayer

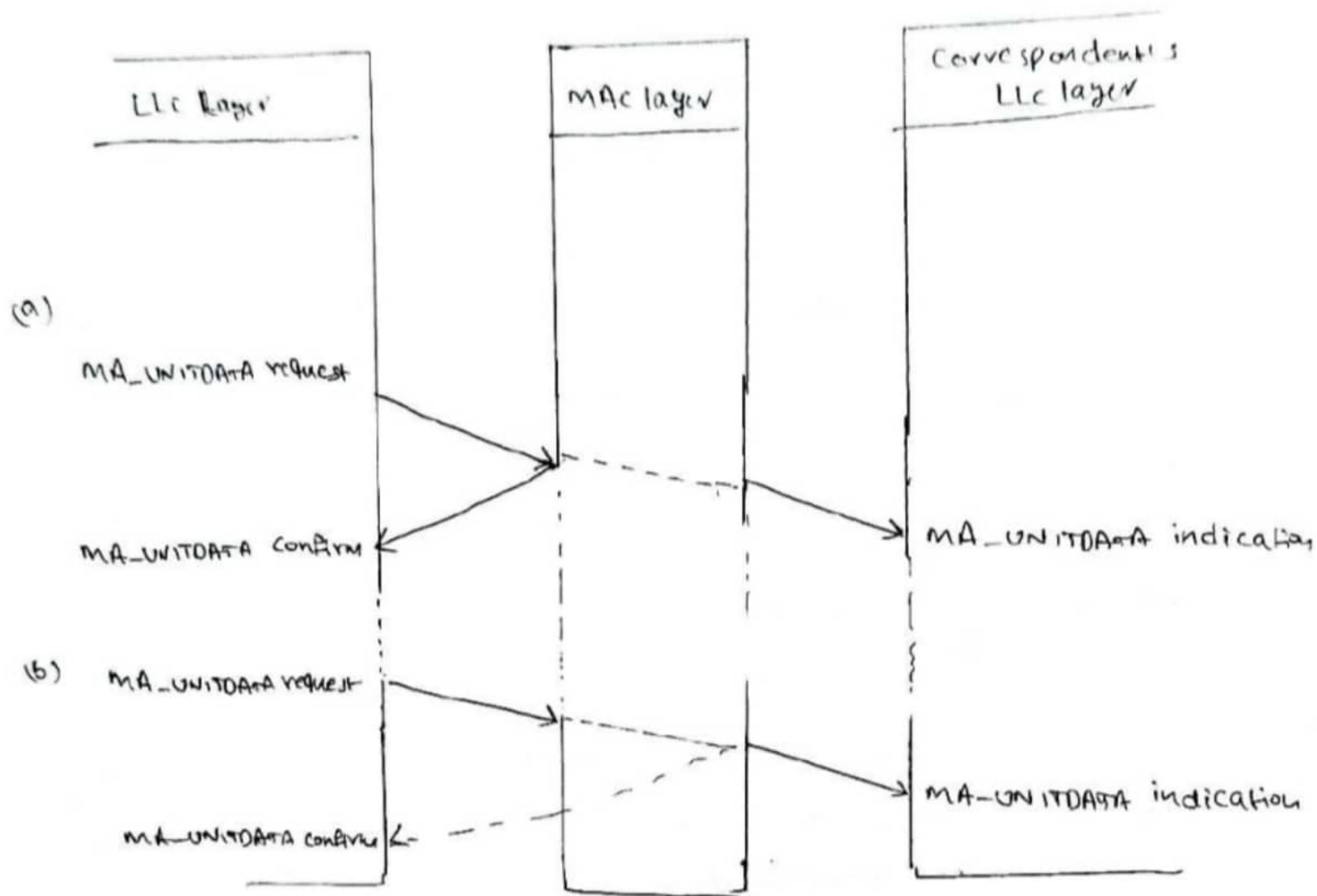
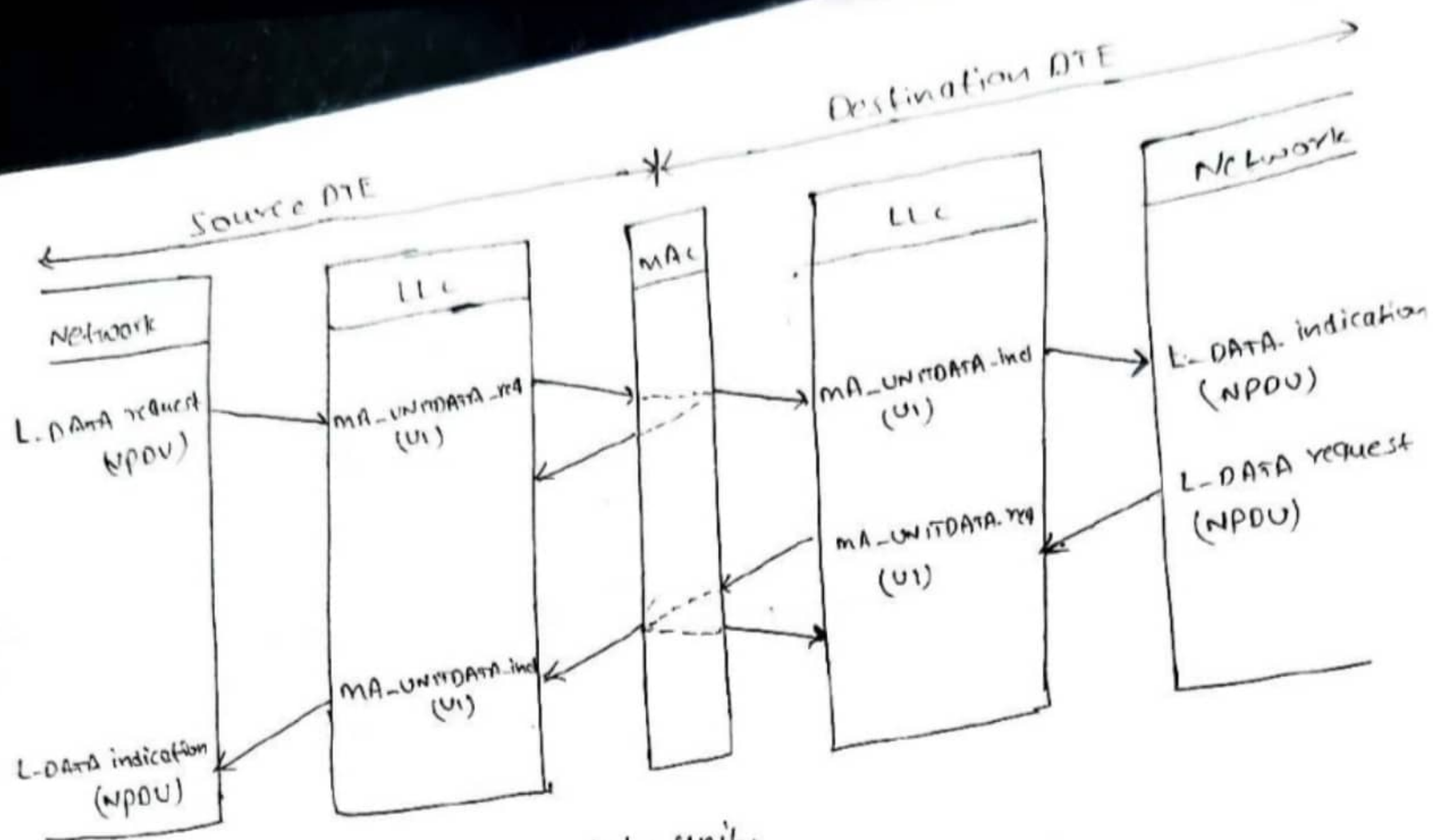


Fig 8.33 : MAC user service Primitives: (a) CSMA/CD ; (b) token ring

→ The LLC ~~sub~~ protocol is based on high-level data link control (HDLC) Protocol which we described earlier. ~~It~~ It therefore supports both a connectionless and connection-oriented mode.

→ when older Ethernet MAC Standard is being used ~~the~~ the LLC sublayer is often not present.

→ The L-UNITDATA request primitive has parameters associated with it. These are : a specification of the source (local) and destination (remote) addresses and the user data (service data unit).



NPDUs = network layer protocol data unit.

Fig: LLC/MAC sublayer interactions.

- Two additional fields are added ^{collectively} what is called the subnet access protocol (SNAP) header. The first is a 3-byte field known as the organisation (org) code - which, with the Internet, all three bytes are set to zero - and a two-byte type field.
- This is same as that used in the original Ethernet Standard and indicates the network layer protocol that created the ~~NPDUs~~ ^{NPDUs} the NPDUs.
- The source and destination addresses are each a concatenation of the MAC sublayer address of the Station and an additional service access point (SAP).

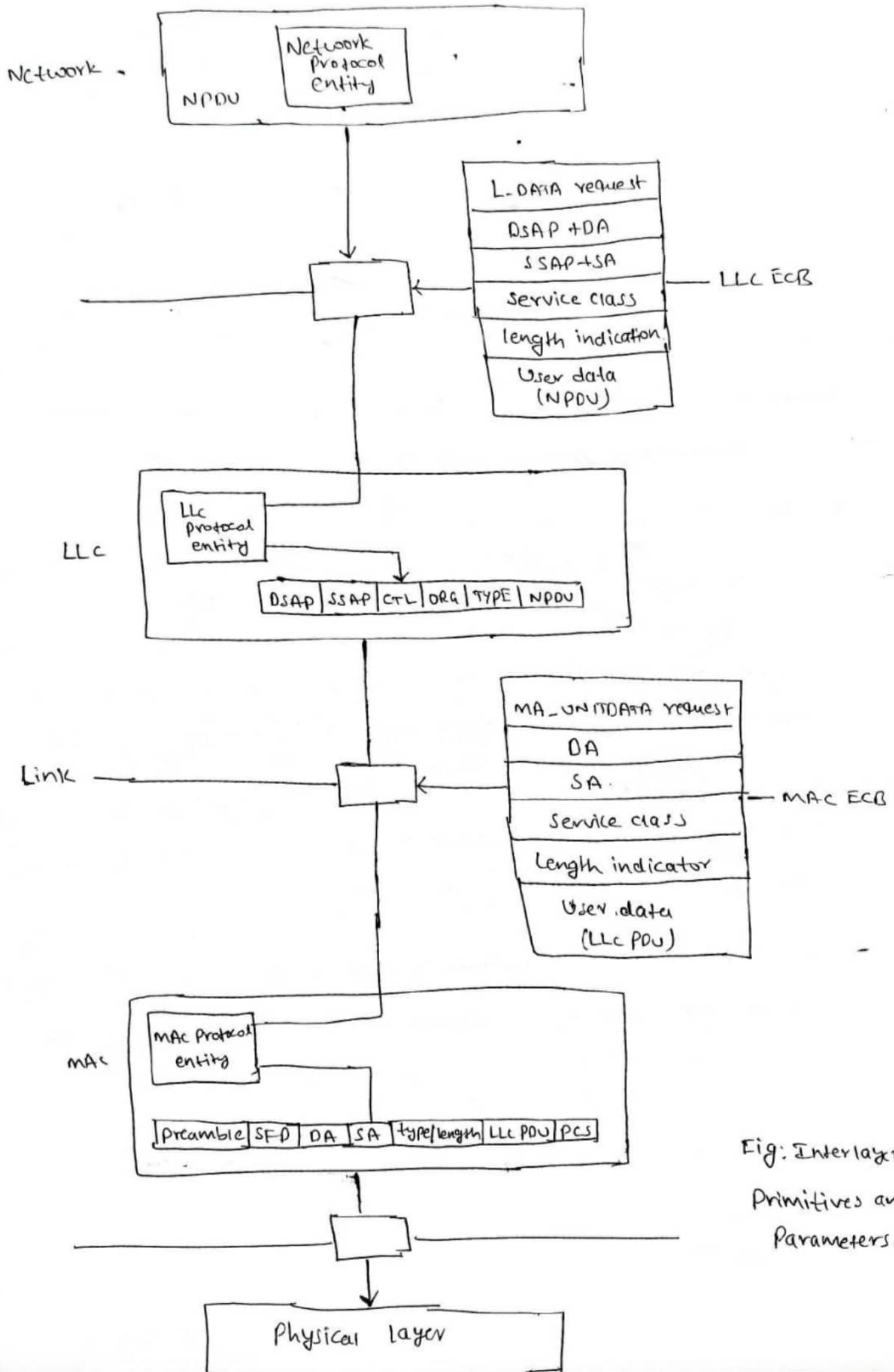


Fig: Interlayer
Primitives and
Parameters.

→ Here the illustration of interactions b/w LLC and MAC sublayers is shown.

→ The LLC sublayer reads the destination and source LLC service ~~access~~ access point addresses (DSAP and SSAP) - from the two address parameters in the event control block (ECB) associated with the LDATA request service primitive - and inserts these at the head of LLC PDU.

→ A similar procedure is followed in NIC of the destination station except that the corresponding fields in each PDU are read and interpreted by each layer.

Network layer

→ The most popular protocol stack used within LANs is Novell Network.

→ The network layer protocol associated with this is a connectionless protocol known as internet Packet Exchange (IPX) protocol and as its name implies, it can route and relay packets over the total LAN.

→ The protocol stack used within the Internet is TCP/IP and the network layer protocol associated with this stack is a connectionless protocol known as the Internet protocol (IP).